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An Automated System for the Condition Grading
of Pokémon Cards

SOCIAL MEDIA DATA ANALYSIS PROJECT
REPORT

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Abstract

This report documents the design, implementation and experimental validation of an automated visual inspection system for collectible trading cards. The system ingests an unconstrained smartphone photo, localizes the card with an oriented object detector, rectifies it removing the perspective distortion, and compares the resulting patch against a memory bank of pristine reference descriptors extracted with DINOv2. The comparison produces a per-patch anomaly distance map that is aggregated into a defect score and mapped onto a consolidated three-tier condition scale (Near Mint/Mint, Good/Excellent, Poor/Played), aligned with the seven-tier Cardmarket reference. The report follows the chronological development of the project: the construction of a synthetic dataset with exact ground-truth geometry, the training and edge-focused fine-tuning of a YOLO11 OBB detector, the introduction of an OpenCV-based physical boundary locking stage, the evolution of the anomaly inspection module from one-shot matching to a photometric few-shot memory bank with spatially constrained nearest-neighbor search, and the final end-to-end evaluation.

Chapter 1

Introduction

The condition of a collectible trading card is the single factor that most determines its market value. Two physical copies of the same Pokémon card can differ in price by an order of magnitude depending on edge whitening, surface scratches, creases or holo foil abrasion. Marketplaces such as Cardmarket grade cards on a seven-tier scale from Mint to Poor, and grading is still done by eye, with all the subjectivity this involves. The same card, photographed under a warm desk lamp or in a dim room, can get different assessments from the same person. This work targets that seven-tier standard with a deliberately consolidated three-tier grading scale (Near Mint/Mint, Good/Excellent, Poor/Played): an implementation choice that matches the resolution of the available anomaly signal, with the fine-grained subdivision left as future work.

The project presented in this report addresses a specific version of this problem. Given a smartphone photo of a single Pokémon card, the system must (i) locate the card and extract it from the background, (ii) compare its surface with an official pristine scan of the same card, and (iii) produce an objective condition grade together with a visual map of the detected defects. The photo conditions we target are deliberately unconstrained: the card may lie on a wooden table or a printed playmat, may be rotated at any angle, and may be photographed under non-uniform lighting.

We identified four technical challenges from the start. The first is *background interference*: cluttered surfaces, arbitrary rotation and protective plastic sleeves all disturb the boundary between card and background. The second is the *confusion between lighting artifacts and physical defects*: specular reflections and cast shadows change pixel values locally in a way that naive defect detectors flag as damage. The third is a geometric detail specific to trading cards: corners are rounded, so any axis-aligned rectangular crop necessarily includes small background wedges near the corners, which then trigger false anomaly detections along the perimeter. The fourth challenge is the *size of the catalog*: tens of thousands of distinct cards exist, which rules out training one dedicated inspection model per card.

The pipeline designed to address these challenges combines an oriented object detector for localization, contrast-limited histogram equalization to handle uneven lighting, and a pre-trained patch-level anomaly detector (AnomalyDINO [3], built on DINOv2 features [2]) for defect localization. Figure 1.1 shows the complete chain: from the raw photograph, through the two quality gates and the geometric stages, down to the anomaly comparison and the final graded output.

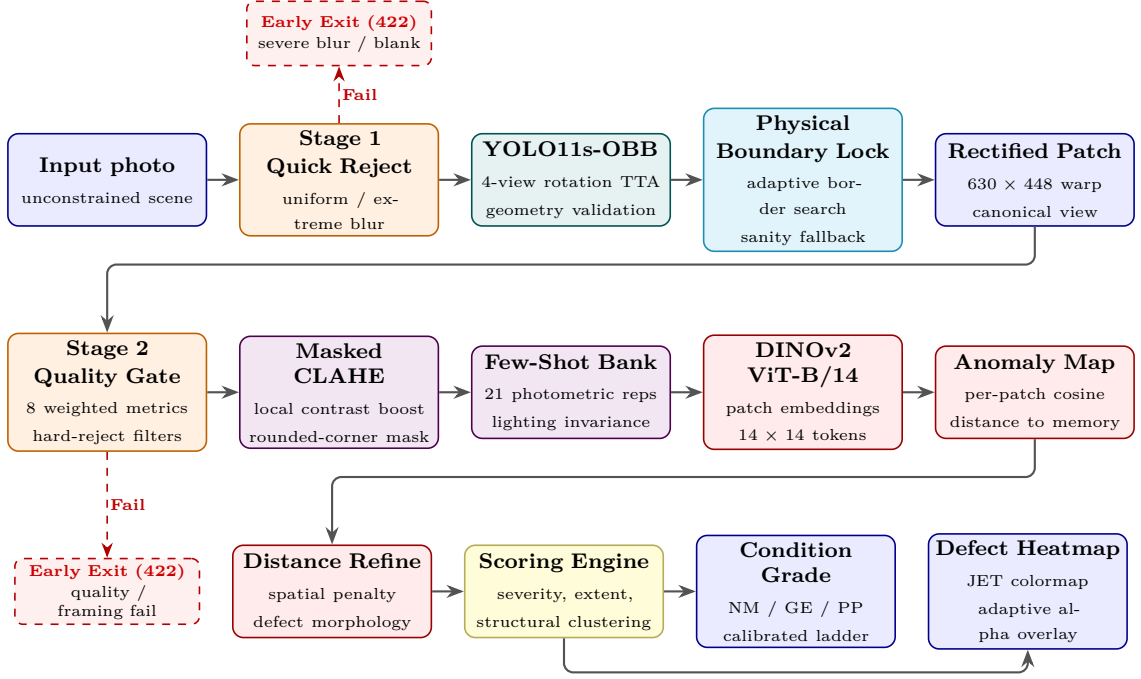


Figure 1.1: The complete inspection pipeline, from input photograph to graded output. Orange blocks represent the two defensive quality gates (with dashed red exits indicating early termination on unusable captures), green the detector with full-card geometry validation, teal the boundary locking stage, violet the photometric preprocessing, red the anomaly inspection chain, and blue the inputs and final outputs.

The document structure follows. Chapter 2 describes the datasets, from the initial prototype dataset of the first training to the final 13,834-card generation run. Chapter 3 describes the components of the pipeline one by one, from input to output, including the choices that did not work. Chapter 4 reports the quantitative and qualitative evidence collected at each stage. Chapter 5 closes with an honest assessment of what the system achieves and where its limits lie.

Chapter 2

Datasets

The project relies on three distinct data assets. Two are datasets of images with completely different roles; the third, the memory bank, is not a dataset at all but a runtime structure rebuilt for every analyzed card. Table 2.1 summarizes them; the rest of the chapter describes how each came to be.

Table 2.1: Overview of the data assets of the project.

Asset		Role	Content
Object Dataset	Detector	trains the YOLO OBB detector	13,834 source card images (13,110 from Hugging Face <code>TheFusion21/PokemonCards</code> [7] plus 724 from the open-source repository <code>1vcian/Pokemon-TCGP-Card-Scanner</code> [8]), rendered into 14,168 synthetic scenes with exact OBB labels (deterministic 70/15/15 card split)
Anomaly Reference Dataset	Detection Cards	searchable catalog for manual reference selection in the API and web interface	20,444 English-language cards indexed from the <code>pokemontcg.io</code> API (local JSON catalog)
Anomaly Bank	Memory	runtime descriptors for the anomaly comparison	30,156 descriptors per analyzed card: 21 representations (canonical scan + 20 photometric variants) $\times$ 1,436 valid patch positions; rebuilt for every analyzed card

2.1 The prototype dataset

The first detector prototype was developed following the synthetic dataset generation strategy introduced by the open-source Pokémon Card Recognition project by 1vcian (<https://github.com/1vcian/Pokemon-TCGP-Card-Scanner>) [8]. It was trained on an initial pool of roughly 700 undamaged digital card scans (specifically 724 cards from the Genetic Apex expansion [8]), pasted onto random background surfaces without systematic defect augmentations. The Nano detector (2.6M parameters) learned that pool well enough to localize clean desk photos, and the failure that ended this first iteration was, on inspection, a data failure rather than a model failure. On heavily played cards with deep horizontal creases the detector split the card into two half-card predictions: the prototype dataset contained no creased card rendered as a single annotated instance, so the network had never been shown that a folded card is still one object. The conclusion was that a model could not learn a pattern absent from its training data, and the dataset had to be rebuilt from scratch: a larger card pool for visual variety, scenes that include worn cards as complete instances, and a leak-free partition for unbiased evaluation.

2.2 Generator rework and expanded dataset

The rebuilt generator produces scenes whose labels are exact by construction, so no manual annotation is needed at any scale. A canonical card image of dimensions (W, H) is placed into a scene through a homography sampled from a continuous distribution over rotation, scale relative to the canvas, and translation constrained to keep the full card inside the frame; the rendered card long side spans 42–66% of the canvas and no card corner may fall closer than 16 pixels to the frame edge. The four source vertices are projected through the homography to obtain the destination quadrilateral, which is normalized to the unit square and written directly into the YOLO OBB label format. No annotation error is possible by design, and the generator can produce arbitrarily many scenes with full control over the difficulty distribution.

Two further mechanisms shape the learning problem. The first is the *hard-negative mix*: about 6.6% of the generated scenes consist of the bare background alone, with the annotation file left empty; the detector is thus actively penalized for firing on texture patterns that resemble cards, which directly suppresses false positives on cluttered desks. The second mechanism is *photorealistic defect augmentation*: before scene rendering, physical wear is procedurally injected into the card texture (structural folds with shadowed troughs and bright fiber highlights, edge chipping that removes the border pigment to expose the white core, linear scratches with a Gaussian spatial profile, specular hotspots, defocus and motion blur). Each defect fires with its own probability (creases 50%, edge whitening 55%, scratches 45%, shadow gradient 45%, specular hotspot 40%, light blur 30%), so most training cards carry several simultaneous wear patterns. The purpose is not to make the detector immune to damage, but to ensure that the training distribution contains cards whose visual boundary remains intact even when the surface is worn, so that detection does not depend on pristine surfaces.

The current Object Detector Dataset expands that initial pool to 13,834 unique source cards. We downloaded 13,110 card scans from the public Hugging Face collection **TheFusion21/PokemonCards** [7] (covering physical card sets across generations) and combined them with the 724 card scans from the open-source Pokémon TCG Pocket repository by 1vcian [8]. All scans were transcoded to WebP at quality 92, reducing the local footprint from over 11 GB of PNGs to about 2.5 GB with no perceptible degradation. Backgrounds come from three sources: 180 tabletop and pattern textures from the DTD (Describable Textures Dataset), distributed via Hugging Face (**Multimodal-Fatima/DTD_partition1_train**); 142 photorealistic CC0 PBR surface materials (wood, marble, fabric, leather, tiles) from the Hugging Face collection **nyuuzyou/cc0-textures**; and 30 curated desk and playmat photos from Unsplash, for a total of 352 background images. The train/validation/test partitioning of the card pool is deterministic, driven by a SHA-256 hash modulo, so that a card appearing in the training split can never appear in a validation or test scene: the zero-leakage property holds for both the source cards and the generated scenes. From these 13,834 source cards, the final generation run produced 14,168 synthetic scenes (Table 2.2): 13,228 positive scenes (placing one card in 88% of scenes and two cards in 12% to train multi-card separation) and 940 hard-negative scenes (6.6%) featuring background textures or unannotated card distractors to suppress false positive detections. Table 2.2 summarizes the split composition.

Table 2.2: Dataset statistics of the final generation run. Source cards are partitioned with a deterministic hash split; hard negatives are scenes with distractor cards that carry no annotation.

	Train	Validation	Test
Scenes	9,921	2,104	2,143
Hard negatives	670	131	139

Total scenes: 14,168; source cards: 13,834; backgrounds: 352; hard-negative ratio: 6.6%.

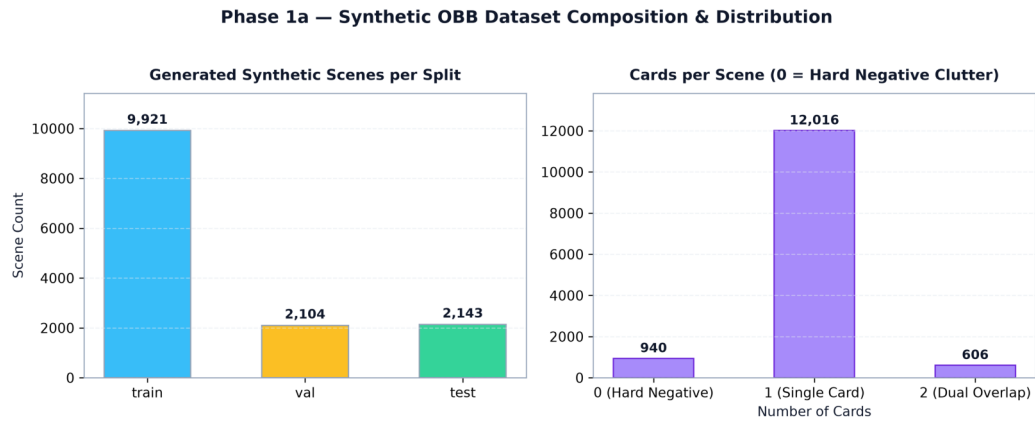


Figure 2.1: Statistics of the generated dataset: scene counts per split, background usage and hard-negative ratio.

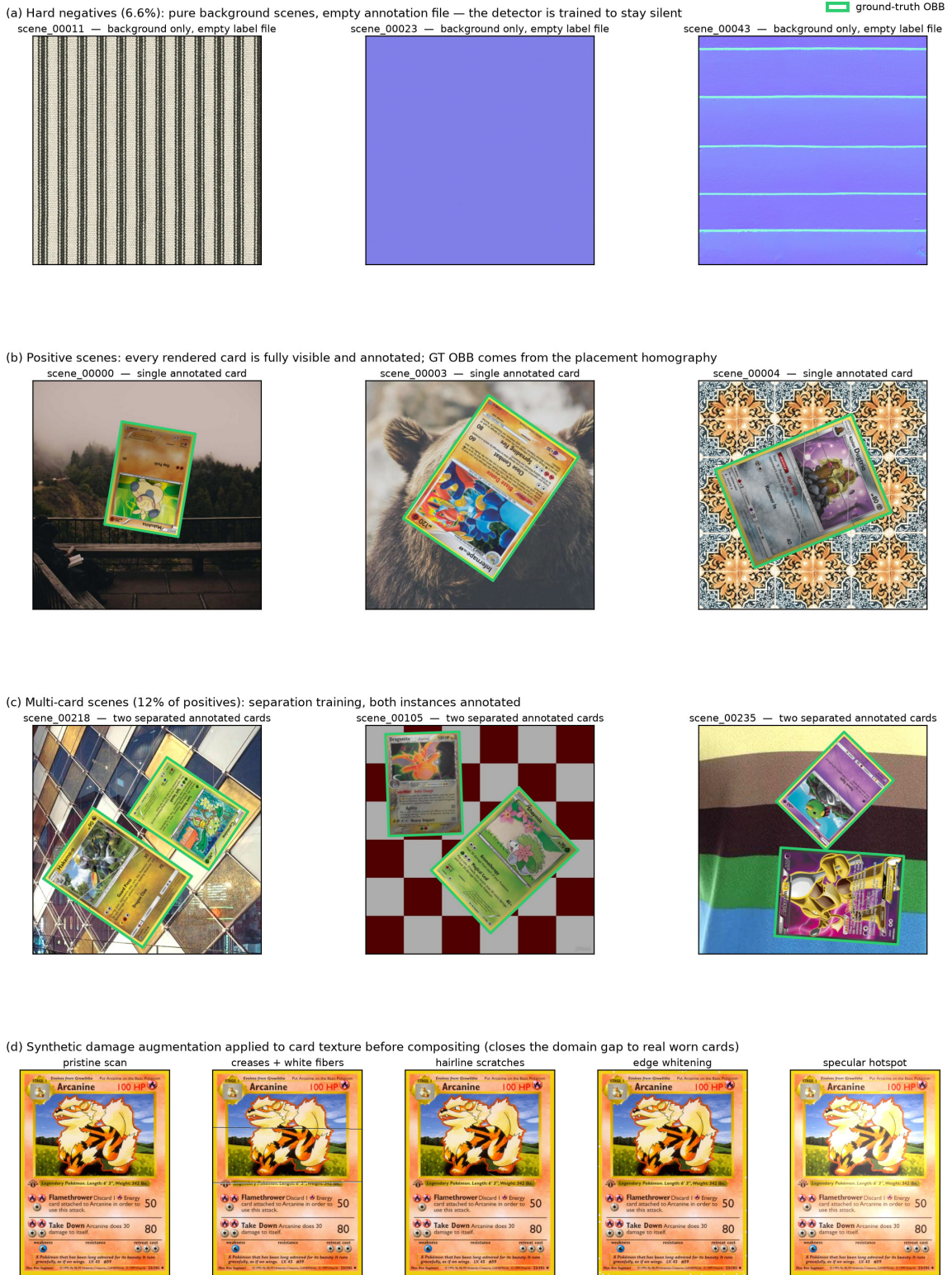


Figure 2.2: Anatomy of the synthetic dataset. (a) Hard-negative scenes contain the bare background with an empty annotation file, teaching the detector to stay silent on card-like textures; (b) every rendered card is fully visible by construction, and its OBB is written directly from the placement homography, so the ground truth is exact; (c) multi-card scenes train instance separation with both instances annotated; (d) synthetic damage augmentation closes the domain gap to real worn cards.

The three assets of Table 2.1 are deliberately kept separate, and the separation is what makes each of them honest. The *Object Detector Dataset* exists to generate scenes with exact geometric

labels for the detector; it never overlaps the evaluation scenes, thanks to the deterministic card split. The *Anomaly Detection Reference Cards Dataset* is the searchable catalog of the deployed interface: the user browses it, selects the pristine scan of the photographed card, and that scan is the only thing the anomaly stage compares against (Section 3.7.2). The system therefore never trains on the cards it inspects: the detector dataset defines what the detector sees, while the reference cards dataset defines what the user can grade.

The *Anomaly Memory Bank* is different in kind: it is not a collection of images but the runtime data structure of the anomaly stage, rebuilt from scratch for every analyzed card. Its size is easy to account for. The canonical scan of the selected card is split into a 45×32 grid of 1,440 patch positions, of which 1,436 fall inside the rounded-corner mask (the 4 corner positions covering background are excluded); the 20 photometric variants are described by the same 1,436 positions each. The bank therefore holds $21 \times 1,436 = 30,156$ descriptors, roughly 93 MB as float32 tensors, and is discarded as soon as a different reference card is selected. Its construction is detailed in Section 3.5.1.

Chapter 3

Proposed Approach

This chapter walks through the pipeline from input to output, examining each component of the chain: for every stage we describe the mechanics introduced and the role it plays in the overall flow.

3.1 The detector: from prototype to edge-focused fine-tuning

The detector is the entry point of the pipeline, and its evolution condenses most of the design lessons of the project. This section retraces that evolution in three steps: the first prototype and the failure that motivated a data-driven redesign, the upgrade to a larger detector supported by a geometry gate, and the edge-focused fine-tuning session that pushed the network’s attention to the card boundary.

3.1.1 First prototype and the fragmentation problem

The first prototype combined a small YOLO11n-OBB detector (2.6M parameters), trained on a small set of undamaged card images, with a one-shot DINOv2 comparator that compared the cropped patch with a single clean scan. It worked on clean desk photos, but on heavily played cards with deep horizontal creases the detector split one card into two half-card predictions with medium confidence, and feeding half-card crops into the comparator produced meaningless defect maps, because the reference scan is a whole card. The cause was in the training data: the prototype dataset contained no worn card annotated as a single object, so the network had never learned that a damaged card is still one object (Section 2). The fix was structural, not a post-processing trick: a new data generator with hard negatives, defect augmentation and full-visibility placement, and a much larger photorealistic training set (Section 2). The rule that remains in production is strict: the detector must return either the complete card or nothing, and broken boxes are discarded upstream.

3.1.2 Object detection: from nano to small

The detector was upgraded from YOLO11n-OBB to YOLO11s-OBB (11.3M parameters) when the dataset reached its final size. Training ran locally with AdamW, a cosine learning-rate schedule starting from 10^{-3} , three warmup epochs, 20 epochs in total, and heavy photometric augmentation (Table 3.1). At inference, four-view rotation test-time augmentation (TTA) runs YOLO on the original photograph and its 90° , 180° , and 270° rotations. The predicted quadrilaterals are mapped back to the original image, detections of the same card are clustered, their cyclic corner orders aligned, and the coordinates fused using confidence and aspect-ratio weights. This reduces orientation-dependent box jitter before the geometric boundary stage. A geometry gate retains candidates with an aspect ratio in $[1.15, 1.65]$ and area of at least 6% of the frame. For tight macro captures, where a card is within 12 pixels of the frame edge or occupies over 55% of

the image, an additional inference on an image padded by 8% using border replication supplies missing boundary context.

Multi-Card Handling and Candidate Selection. In real-world captures, multiple cards may be present in the same photograph (such as two cards placed side by side or a neighboring card partially visible on the desk). The system manages multi-card scenes systematically:

1. **Independent Localization:** Non-Maximum Suppression (NMS) separates distinct cards as independent oriented bounding boxes whenever their centers are spaced apart (distance $> 35\%$ of the diagonal). This prevents adjacent cards from mutually suppressing each other.
2. **Automated Geometric Candidate Ranking:** When multiple cards are detected, the pipeline automatically ranks candidates to select the primary card for grading. The ranking balances YOLO detection confidence with a geometric penalty for aspect ratio deviation:

$$\text{Score} = \frac{\text{conf}}{1.0 + 3.0 \cdot |\text{AR} - \text{AR}_{\text{target}}|} \quad (3.1)$$

where $\text{AR} = \max(w, h) / \min(w, h)$ is the bounding box aspect ratio and $\text{AR}_{\text{target}} \approx 1.40$ (88×63 mm standard Pokémon card format). Candidates whose aspect ratio deviates significantly (such as cards partially cut off at the image border or background artifacts) are penalized, ensuring that the fully framed, central card is chosen automatically.

Table 3.1: Training protocol for YOLO11s-OBB on the synthetic dataset (MPS backend, input 640×640).

Parameter	Value
Epochs / batch	20 / 16
Optimizer	AdamW, cosine LR
lr_0 / warmup	10^{-3} / 3 epochs
Patience / weight decay	8 / $5 \cdot 10^{-4}$
Mosaic / close-mosaic	1.0 / 5 epochs
Mixup / erasing	0.15 / 0.20
Rotation / flip	180° / 0.5
Scale / translate / perspective	0.5 / 0.1 / 0.0005

3.1.3 Edge-focused fine-tuning

After the main training, the residual error source was corner precision itself. At 640×640 resolution, YOLO’s finest feature map has a stride of 8 pixels, which puts a floor under the achievable corner accuracy; in addition, the ProbIoU gradient decays rapidly once the IoU exceeds 95%, and the classification loss, already saturated at 100% accuracy, competed with localization for gradient magnitude. The OBB training objective is a weighted sum of four terms:

$$\mathcal{L}_{\text{OBB}} = \lambda_{\text{box}} \mathcal{L}_{\text{box}} + \lambda_{\text{cls}} \mathcal{L}_{\text{cls}} + \lambda_{\text{dff}} \mathcal{L}_{\text{dff}} + \lambda_{\text{angle}} \mathcal{L}_{\text{angle}},$$

where $\mathcal{L}_{\text{box}}$ is the ProbIoU regression on the rotated box, $\mathcal{L}_{\text{cls}}$ the BCE classification term, $\mathcal{L}_{\text{dff}}$ the distribution focal loss on the box-edge coordinates and $\mathcal{L}_{\text{angle}}$ the orientation term. The initial training employed the standard default loss weights provided by the Ultralytics YOLO11-OBB framework ($\lambda_{\text{box}} = 7.5, \lambda_{\text{dff}} = 1.5, \lambda_{\text{angle}} = 1.0, \lambda_{\text{cls}} = 0.5$). A dedicated five-epoch fine-tuning session from the best checkpoint rebalanced the weights toward edge precision: $\lambda_{\text{box}} 7.5 \rightarrow 18.0$, $\lambda_{\text{dff}} 1.5 \rightarrow 3.5$, $\lambda_{\text{angle}} 1.0 \rightarrow 2.5$, $\lambda_{\text{cls}} 0.5 \rightarrow 0.15$, with the learning rate lowered to $3 \cdot 10^{-4}$ and mosaic and mixup disabled so that no artificial edge cutoffs were shown to the model. This fine-tuning session operates on the same 10,000-scene synthetic dataset as the main training,

preserving the validation split; the improvement stems from re-weighting the loss surface and removing destructive mosaic cuts, rather than expanding the training data. Table 3.2 lists the re-weighting relative to the Ultralytics defaults.

Table 3.2: Loss weights λ for the baseline training (standard Ultralytics defaults) and the edge-focused fine-tuning. The ratios quantify how much each term was emphasized relative to the default protocol.

Weight	Ultralytics Default	Fine-tuning	Ratio	Purpose
λ_{box} (ProbIoU)	7.5	18.0	$\times 2.40$	Corner precision
λ_{dfl}	1.5	3.5	$\times 2.33$	Edge-coordinate sharpness
λ_{angle}	1.0	2.5	$\times 2.50$	Orientation accuracy
λ_{cls}	0.5	0.15	$\times 0.30$	De-emphasize solved task

3.2 Physical Boundary Lock

While oriented bounding box detection yields high mean Intersection-over-Union, convolutional feature downsampling ($\times 32$ stride) and bounding box regression leave small boundary uncertainties (typically 3 to 8 pixels). In visual inspection tasks, even a slight corner drift introduces perspective distortion into the rectified crop (Section 3.2.4), warping the 14×14 patch grid fed to DINOv2 and triggering spurious anomaly detections along the printed frame. In addition, when cards are photographed inside protective sleeves or held in hand, the detector can latch onto the outer transparent plastic margin rather than the cardstock itself.

The *Physical Boundary Lock* (PBL, implemented in `_lock_physical_card`) refines the four-view YOLO consensus before rectification. Its color estimator is self-calibrating in CIE-Lab space: no card-border hue is predefined. It therefore applies the same procedure to yellow vintage frames, silver borders, dark frames, and other printed perimeter colors. PBL consists of three deterministic stages:

3.2.1 Step 1: Canonical Aspect Ratio Regularization

The four TTA-fused YOLO corners are converted into their minimum-area bounding rectangle (`minAreaRect`), preserving the detected center and orientation while removing residual quadrilateral skew. Its dimensions are projected onto the standard card aspect ratio ($\rho_{\text{canon}} = 88/63 \approx 1.397$), followed by a calibrated 1.5% shrink along the long axis. The resulting quadrilateral $\mathbf{C}_{\text{reg}}$ is the conservative fallback and reference for subsequent corrections.

3.2.2 Step 2: Adaptive Border Estimation and Contour Sweep

The border color is estimated from the image in CIE- $L^*a^*b^*$ space rather than selected from a predefined palette:

1. **Annulus Sampling:** A ring around $\mathbf{C}_{\text{reg}}$ (6 px outside to 18 px inside) supplies possible border pixels; a separate ring 8–30 px outside estimates the surrounding background color.
2. **Color Clustering:** OpenCV k -means ($k = 6$) proposes dominant Lab colors from the annulus.
3. **Perimeter Validation:** For each color, thin strips are tested at several inward depths along all four sides. A candidate needs at least 35% coverage on each side and 50% mean coverage; it is then scored by perimeter support and contrast from the outer background.
4. **Similarity Map and Contour Search:** The selected color defines a soft Lab-distance likelihood map. A 20-threshold sweep (percentiles 40–98), followed by an 11×11 morphological closing, extracts candidate quads. A candidate must have contour area between 90%

and 100% of the YOLO box, minimum-rectangle area between 88% and 100%, short side of at least 80 px, aspect-ratio deviation below 0.040 from 88:63, and center displacement below 2% of the box long side. Among surviving candidates, the closest aspect ratio wins, with larger area breaking ties.

3.2.3 Step 3: Sanity Gate and Fallback

The final sanity gate rejects corrections farther than $\max(10 \text{ px}, 2.5\% \text{ of the box long side})$ from $\mathbf{C}_{\text{reg}}$. Together with the non-expanding area and center constraints, this rejects outer background contours (for example, the apparent light frame around a card on a white surface). If no candidate passes all tests, PBL returns $\mathbf{C}_{\text{reg}}$ unchanged.

The gate is intentionally conservative: it can tighten a supported perimeter candidate but cannot expand the YOLO detection onto a larger background rectangle. On 455 held-out validation scenes, the complete four-view TTA+PBL pipeline achieves a mean corner error of 1.380 px and mean polygon IoU of 98.80% (Section 4.2.1).

3.2.4 The rectified patch

The rectified patch is the standard form in which the card leaves the localization chain and enters every downstream stage: the quality gate scores it, masked CLAHE equalizes it and the anomaly inspection compares it with the reference. It is produced by a final warp step (implemented as `warp_card`): the four corners delivered by the Physical Boundary Lock are first put in a consistent top-to-bottom order, then `getPerspectiveTransform` computes the perspective transform that maps them onto the four corners of a fixed 630×448 upright rectangle, and `warpPerspective` resamples the photo accordingly. The name “rectified” refers exactly to this effect: the quadrilateral detected in the scene – tilted, rotated and viewed in perspective – is un-distorted into a frontal view of the card, in which the 88:63 proportions are restored and the image content is axis-aligned, upsidedown-free and free of perspective skew. The fixed output size makes every patch directly comparable with every other patch and with the reference scan: same geometry in, same geometry out, so that a difference between query and reference descriptors can only come from the card surface, not from how the photo was framed.

3.3 Two-stage quality gate

The quality gate is the module that filters out photographs that are genuinely unusable—frames so degraded that computing an anomaly score on them would be meaningless. It acts as an upstream trigger that prevents the detector and the anomaly inspection from executing on hopeless inputs, saving computational resources and eliminating false defect reports. The module is configured via `configs/default.yaml` and organized in two cascading stages: Stage 1 operates on the raw image before YOLO inference, while Stage 2 operates on the rectified YOLO crop.

3.3.1 Stage 1: Quick Reject (Raw Image)

Stage 1 is a lightweight pre-check that executes on the uncompressed raw photograph before neural inference. It enforces three deterministic boundary conditions:

1. **Minimum Resolution Floor:** The image must measure at least 300×300 pixels (`min_width`, `min_height`) to filter out corrupted thumbnails and low-resolution assets.
2. **Near-Uniformity Gate:** The grayscale pixel standard deviation must satisfy $\sigma_{\text{pixel}} \geq 10.0$ (`min_pixel_std`). Completely black, white, or corrupted frames are immediately discarded.

3. **Catastrophic Defocus Gate:** The global Laplacian variance must satisfy $\text{Var}(\Delta I) \geq 15.0$ (`min_laplacian_var`). This threshold is kept intentionally permissive to account for smooth desk surfaces that naturally produce low gradient responses.

3.3.2 Stage 2: Detailed Quality Gate (Crop Assessment)

Photographs passing Stage 1 are passed to the YOLO11s-OBB detector. Once the oriented card quadrilateral is extracted and rectified, Stage 2 evaluates the card crop across eight distinct quality metrics, each normalized to the unit interval $[0, 1]$. The individual metrics are aggregated into a composite quality score:

$$Q = \sum_{i=1}^8 w_i \cdot m_i, \quad (3.2)$$

where $\sum w_i = 1.0$. In addition to the aggregate pass threshold ($Q \geq 0.35$), each metric is guarded by an independent **hard-reject threshold**. If any metric falls below its hard-reject boundary, the acquisition is rejected immediately regardless of how high the aggregate score may be. Table 3.3 details the complete configuration parameterized in `configs/default.yaml`.

Table 3.3: Quality gate Stage 2 metrics, weights (w_i), hard-reject thresholds, and operational definitions as configured in `configs/default.yaml`.

Metric	Weight w_i	Hard Reject	Operational Definition / Target
Sharpness	0.35	$s < 0.15$	Normalized Laplacian score; tolerates low-texture holo foil
Exposure	0.20	$s < 0.25$	Normalized score on clipped tails and luminance balance
Noise	0.15	$s < 0.10$	Normalized score from residual Gaussian filter variance
Blown-out Glare	0.10	> 0.50	Glare coverage fraction in HSV space ($V \geq 240, S \leq 40$)
Resolution	0.10	$< [448, 630]$	Minimum physical crop dimensions for DINOv2 ViT-B/14
Detection Confidence	0.05	< 0.12	YOLO OBB confidence (aligned with inference threshold)
Aspect Ratio	0.03	$s < 0.30$	Score penalizes deviation from 0.714 (63/88); tolerance $\pm 30\%$
Relative Area	0.02	$s < 0.45$ ($a_{\text{rel}} < 15\%$)	Normalized score $s_{\text{area}} = \min(3a_{\text{rel}}, 1.0)$; rejects cards occupying $< 15\%$ of frame
Aggregate Pass	1.00	$Q \geq 0.35$	Minimum weighted composite score to proceed downstream

Two calibration decisions in Table 3.3 deserve special emphasis:

- **Framing and Relative Area ($< 0.45 \iff < 15\%$):** When a card occupies a tiny fraction of the photographic frame, perspective homography must heavily upscale the raster patch to reach the canonical 630×448 grid. This artificial upsampling introduces interpolation blur that corrupts high-frequency DINOv2 feature descriptors. The area score function is parameterized as $s_{\text{area}} = \min(3 \cdot a_{\text{rel}}, 1.0)$; enforcing a hard-reject threshold of 0.45 mathematically requires $a_{\text{rel}} \geq 15.0\%$, cleanly rejecting small, distant cards.
- **Tolerant Sharpness Threshold (0.15):** Catastrophic optical blur is already intercepted by Stage 1. In Stage 2, setting the crop sharpness threshold too aggressively would penalize legitimate holographic cards, which feature expansive smooth foils and minimal high-frequency edge texture. Calibrating the hard reject to 0.15 ensures that reflective cards pass safely while retaining strict bounds on out-of-focus captures.

Acquisitions failing either stage return an explicit HTTP 422 Unprocessable Entity error accompanied by a detailed diagnostic payload listing the exact triggering failure conditions.

3.4 Masked CLAHE and the rounded-corner mask

Illumination robustness is handled by contrast-limited adaptive histogram equalization (CLAHE, [4]). Histogram equalization is a remapping of intensity values: for each gray level, a histogram counts how many pixels have that value, and the cumulative distribution of these counts defines a transfer function that maps each input intensity to a new one. Under this mapping, well-populated gray levels — where the image actually has detail — get stretched apart, while empty regions of the range get compressed, so the luminance range is used more evenly and details emerge in both dark and overexposed areas. Doing this globally helps little when illumination varies across the photo, so CLAHE works locally: the image is split into an 8×8 grid of tiles and each tile is equalized against its own histogram, adapting to local lighting. The “contrast-limited” part guards against noise amplification: the clip limit of 2.0 is a cap on the height of the histogram bins — no bin may exceed twice the height it would have if the tile’s pixels were spread uniformly over all bins. A bin that exceeds the cap is clipped and its surplus votes are redistributed over all the other bins, so no single dominant gray level can monopolize the contrast stretch (which is what turns sensor noise into visible artifacts in flat regions). Neighboring tiles are then blended by bilinear interpolation to avoid visible seams. The transform acts only on the L channel of the LAB color space, so contrast is enhanced while hue and saturation, the chromatic identity of the artwork, are left untouched.

The equalization is confined to the card interior by the same mask used for feature extraction, which matters more than it sounds: masked-out pixels neither contribute to the local histograms nor get brightened. If the dark background outside the card participated in the statistics of the border tiles, border pixels would be equalized toward the background’s dark levels, brightening the border and diluting its defects against this artificial contrast. With the mask, every border tile’s tonal statistics describe card content only, so the border keeps its true appearance.

The mask itself exists because of the rounded-corner problem listed in the introduction. A rectangular crop of a card necessarily contains four small background wedges at the corners; these wedges would appear to the anomaly detector as glaring defects with no physical counterpart. The mask is a rounded rectangle whose corner radius is 3.5% of the card width, eroded by 1% on the perimeter (about 4 pixels per side) and feathered with a 5-pixel Gaussian, so that the crop edge and the background wedges are excluded from every downstream computation while the printed card border is preserved. An ablation of this component is reported in Section 4.4.3.

3.5 Anomaly inspection with DINOv2

The inspection core follows the AnomalyDINO approach [3]. The rectified 630×448 patch is resized to the DINOv2 ViT-B/14 input convention and split into 14×14 patches, yielding a 45×32 grid of 1,440 positions, of which 1,436 fall inside the rounded mask. DINOv2 [2] produces a 768-dimensional ℓ_2 -normalized descriptor per position. Each test descriptor is matched against the memory bank of the reference card, and the one-minus-cosine distance of the best match forms the anomaly distance map. The image-level score aggregates the top 1% of distances (`mean_top1p`), the standard aggregation of AnomalyDINO.

The one-shot formulation, with a single official scan in the memory bank, produced maps full of false alarms on real photographs. Warm tungsten lighting shifted the whole color distribution, sensor grain at high ISO appeared as abrasion, and the holo foil produced distance spikes when the viewing angle changed. Each of these effects is photometric, not physical, and the system needs to absorb them without losing sensitivity to real scratches.

3.5.1 The few-shot memory bank of camera variants

Collecting many real photographs of every catalog card was out of the question, so we generate the variability synthetically. The memory bank is built per card on demand: when the user selects a reference card, the backend retrieves its canonical scan (either from the local disk cache or downloaded via HTTP on first access) and derives from it 20 deterministic camera-effect variants that imitate what smartphone cameras do in difficult conditions. The variants include warm tungsten and cool fluorescent white balance, ISO sensor noise ($\sigma = 14$), lens defocus, motion blur, vignetting, a flash hotspot, JPEG compression at quality 60, ISP smoothing, a harsh phone shadow gradient, lateral window light, under-exposure (-22% luminance), a three-step over-exposure ladder ($+22\%$, $+45\%$ and $+70\%$ luminance), high- and low-contrast tone curves, boosted saturation, chromatic aberration and a periodic holo sheen.

The bank therefore contains 21 representations of the selected card (the canonical scan plus 20 variants), each described by 1,436 valid patch descriptors, for a total of $21 \times 1,436 = 30,156$ descriptors. The arithmetic is simple: the patch grid has $45 \times 32 = 1,440$ positions and the rounded-corner mask removes the 4 positions that cover the background wedges, leaving 1,436 valid ones per representation.

The initial latency observed when loading a card for the first time is due to this extraction step: passing all 21 high-resolution representations through the DINOv2 vision transformer takes roughly 2–3 seconds. However, this cost is paid only once: the extracted descriptor tensors are permanently cached to disk (as serialized binary arrays). On subsequent sessions, the system achieves an immediate cache hit, loading the descriptors directly into GPU/RAM in milliseconds without re-running the neural backbone. During inference, each test patch is exposure-matched to the reference luminance before scoring, compared against the full bank in all four in-plane orientations, and the best match automatically absorbs harmless lighting and rotation shifts. The 20 variants of the Dark Charizard reference used in the experiments are shown in Figure 3.1.

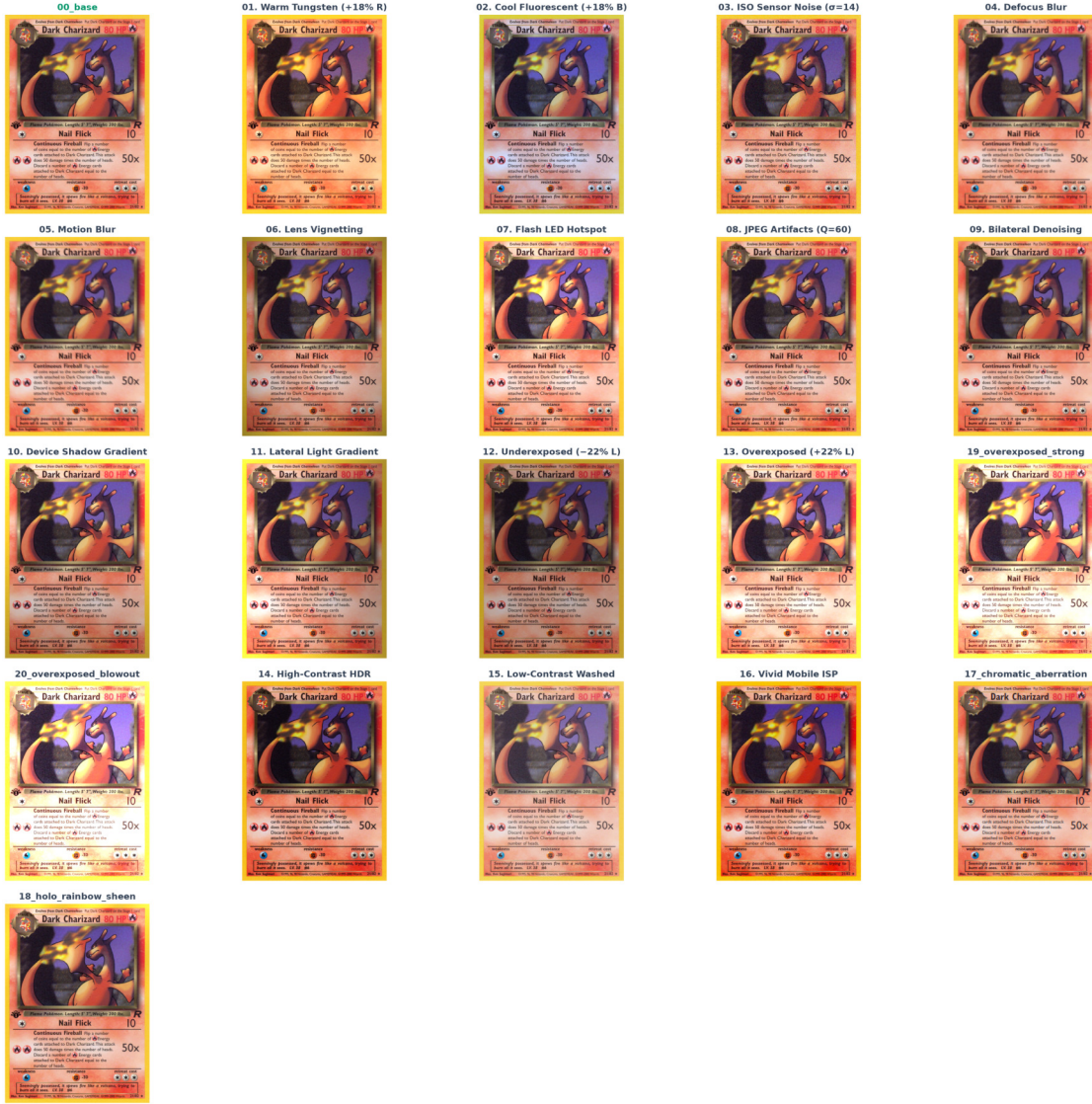


Figure 3.1: The full few-shot bank: the canonical base plus the 20 camera-effect variants generated from a single scan. Each variant is deterministic and imitates one smartphone acquisition condition.

A dual-bank arrangement keeps the two operating modes available: the one-shot bank (1,436 descriptors) and the few-shot bank ($21 \times 1,436$). Both reside pre-normalized on the accelerator, and switching between them at render time costs hundreds of milliseconds, with no re-run of the backbone.

3.5.2 Spatially constrained nearest-neighbor matching

Trading cards feature many repetitive visual elements, such as identical energy symbols, text characters, mana icons, and border segments. If we perform an unconstrained search across the entire card, a damaged query patch (e.g., a scratched energy symbol at the top of the card) could easily find an undamaged identical symbol located at the bottom of the reference card. The model would report a near-zero anomaly distance, completely concealing the physical defect.

To prevent this shortcut, the nearest-neighbor search is restricted to the correct physical neighborhood: a reference patch is only considered a valid candidate if it lies close to the query patch’s expected position on the card.

The effective similarity is simply the visual feature similarity adjusted by a soft spatial penalty based on the physical grid distance d between the two patches:

$$S_{\text{effective}} = \cos(\mathbf{f}_q, \mathbf{f}_r) - \text{Penalty}_{\text{spatial}}(d) \quad (3.3)$$

The spatial penalty follows three intuitive rules:

- **Zero penalty within tolerance ($d \leq 3$ patches):** A small tolerance radius allows minor homography inaccuracies and camera perspectives to match freely without penalty.
- **Gradual penalty ($3 < d < 6$ patches):** As the distance increases, the penalty grows linearly, progressively discouraging distant candidates.
- **Maximum penalty ($d \geq 6$ patches):** Matches located far away or in different quadrants of the card receive the maximum penalty ($w_{\text{spatial}} = 0.45$), guaranteeing that a defect in one region cannot be masked by healthy graphics from another.

3.6 Condition scoring

Once the patch-level feature comparisons are computed, the system translates the resulting 2D anomaly distance map into a condition grade. While the heatmap shows where defects are located on the card, appraisal also requires a single numerical score that reflects the physical condition and maps onto standard collector tiers (such as Near Mint, Excellent, or Played). This section explains how the anomaly map is converted into a composite defect score and categorized into condition grades.

3.6.1 From anomaly map to composite defect score

The feature matching stage produces a 2D anomaly distance map over the 1,436 valid patch positions of the rectified card grid (45×32 patches, excluding the rounded corner background areas). Each patch value represents the cosine distance between the query patch descriptor and the closest matching patch in the reference memory bank.

To convert this map into an accurate condition score, a naive global average is not suitable: a severe scratch on an otherwise mint card would be completely lost if averaged over all 1,436 patches. Conversely, taking only the single worst patch value would make the score hypersensitive to a stray dust particle or minor camera noise.

To balance sensitivity and stability, the final composite score S_{comp} combines three straightforward, complementary terms:

$$S_{\text{comp}} = S_{\text{peak}} + w_{\text{area}} \cdot A_{\text{defect}} + \Delta_{\text{morph}} \quad (3.4)$$

where:

1. **Peak defect severity (S_{peak}):** The average distance of the worst 1% of patches (the top 14 patches). This reliably captures localized physical damage—such as deep scratches, ink abrasions, or creases—without over-reacting to a single isolated outlier pixel.
2. **Defect surface area (A_{defect}):** The percentage of the card surface whose patch distances exceed the defect threshold ($\tau = 0.52$). Scaled by a moderate weight of $w_{\text{area}} = 0.15$, this term penalizes widespread surface wear (such as broad scuffing or dulling of the card) while preventing benign reflective highlights from artificially inflating the score.
3. **Structural morphology penalty (Δ_{morph}):** Using connected component analysis on the defective patches, the system differentiates coherent physical damage from random image noise. Isolated noisy patches incur no penalty, but small extra increments are added if defects form:

- *Large clusters*: connected defect regions larger than 25 patches, representing significant contiguous damage (+0.005 per excess patch).
- *Linear scratches*: elongated defect components (aspect ratio > 4.0), representing continuous scratch lines (+0.010 per detected line).

This combination accounts for both the intensity of localized flaws and the spatial extent of wear across the card.

3.6.2 Condition classification and Cardmarket tiers

To provide an intuitive appraisal, the continuous composite score is mapped onto a standardized condition tier system aligned with the established Cardmarket convention (Table 3.4). While commercial numismatic standards specify seven discrete grades ranging from Mint to Poor, distinguishing between adjacent categories in the heavily worn spectrum (such as Light Played versus Played) requires subjective human discretion that exceeds the noise margin of uncalibrated mobile captures.

To improve reproducibility and stability, the system projects the seven-tier scale onto three consolidated preservation tiers:

- **Near Mint/Mint (NM)** (Score < 0.450): Covers cards exhibiting negligible handling, retaining crisp factory sheen and sharp edges. The upper boundary (0.450) comfortably accommodates immaculate specimens photographed under standard domestic smartphone lighting.
- **Good/Excellent (GE)** ($0.450 \leq \text{Score} < 0.750$): Captures cards with visible but minor play wear, faint surface scuffs, slight perimeter whitening, or benign reflective foil sheens.
- **Poor/Played (PP)** (Score ≥ 0.750): Encompasses cards with heavy, structurally evident deterioration, including creases, deep scratches, or extensive foil peeling.

In addition to the three-tier categorization, a binary defect verdict is emitted using a decision threshold of 0.620, separating cards in collectible condition from those with notable physical anomalies.

Table 3.4: Consolidated condition bands used by the classifier, mapping the continuous composite score onto standardized commercial preservation tiers.

Grade	Composite score range	Equivalent Cardmarket tiers
Near Mint/Mint (NM)	< 0.450	Mint, Near Mint
Good/Excellent (GE)	$[0.450, 0.750)$	Excellent, Good
Poor/Played (PP)	≥ 0.750	Light Played, Played, Poor

3.6.3 Language support limitation

The local reference database is currently indexed on 20,444 English-language card scans. Although multilingual queries were initially explored, remote catalog querying proved vulnerable to network timeouts and inconsistent metadata indexing. Consequently, the inspection pipeline operates strictly against canonical English reference scans.

This constraint carries practical implications for foreign-language cards (such as Italian, Japanese, or German prints). Because the DINOv2 vision transformer extracts dense representations of printed typography, differences in fonts and wording between query and reference generate localized feature distances in the text bounding box (see Section 4.6.3). For non-English cards, this textual divergence acts as an upward bias on the composite score, occasionally shifting an unworn card into a lower condition tier. Extending the memory bank to multilingual printings is discussed in Section 5.1.

3.7 Output rendering and reference selection

The pipeline pairs a defect visualization engine with a catalog retrieval flow. This section describes the image processing used to produce the defect heatmaps, and details the manual selection protocol that provides the pristine reference baseline for anomaly comparison.

3.7.1 Heatmap rendering

The distance map is turned into a readable overlay in several steps. A dead zone hides the normal noise floor; gamma correction ($\gamma = 2.0$) re-maps the remaining distances, suppressing faint variation and amplifying real defects; the JET colormap colors the result, and a bilateral filter (diameter 9, $\sigma_{\text{color}} = 75$, $\sigma_{\text{space}} = 75$) smooths the colormap noise while keeping defect edges sharp. The overlay opacity adapts to the severity of the analysis: the alpha ceiling is 0.40 when the mean normalized anomaly is low, 0.55 in the mid range and 0.70 when the card shows extensive damage, so a mostly clean card stays visible under a light overlay, while a damaged one gets a stronger visual emphasis.

3.7.2 Manual reference selection

The current implementation requires the user to select the reference card from the catalog before analysis. To eliminate network latency, card lookup runs fully offline against a local pre-indexed database of 20,444 cards, allowing real-time text filtering by card name (e.g., “Charizard-EX”) or set numbering (e.g., “100/106”). When the user confirms a card, the system checks whether its official scan is already cached on disk; if missing, a single HTTP call downloads the high-resolution asset from the remote CDN and archives it locally. Subsequent selections of that card read the image and its memory bank directly from local storage, ensuring the system operates with zero external network dependencies once cards are cached. An automated visual retrieval system based on image embeddings was prototyped but is left for future work.

Chapter 4

Experiments

This chapter presents the quantitative evidence gathered throughout development, organized by pipeline component so that related analyses are grouped together: the card detector, the geometric refinement performed by the Physical Boundary Lock (PBL), the acquisition quality gate and preprocessing stage, the anomaly scoring module, and finally the full-system evaluation. The datasets underlying these experiments are described in Chapter 2; the results below quantify how the components built on them perform.

4.1 Detector evaluation

Accurate card localization and perspective rectification are required before downstream processing: residual angular tilt, aspect ratio distortion, or bounding box dilation directly misaligns the patch grid fed to the DINOv2 backbone. This section evaluates the oriented detection subsystem (YOLO11s-OBB) across its development stages:

1. Architecture scaling and unified training progression across both the initial 10-epoch base phase and the subsequent 5-epoch edge-focused fine-tuning phase;
2. Rotational invariance across 360° in-plane orientations and validation predictions across diverse scenes;
3. Multi-card clutter separation to verify independent object localization.

4.1.1 Architecture scaling and unified training progression

Card localization began with an exploratory prototype and scaled up to a production-grade oriented bounding box (OBB) model. Three detector configurations were trained and compared on the validation split: an initial lightweight YOLO11n-OBB prototype (trained on a preliminary set of 1,000 clean scenes), an intermediate YOLO11s-OBB model trained on an early subset of 3,000 synthetic scenes, and the final production YOLO11s-OBB model trained on the full synthetic dataset of 10,000 scenes (`pokemon_obb`) across 10 base epochs and subsequently refined with 5 edge-focused fine-tuning epochs.

The dataset remains identical between the base training and the fine-tuning of YOLO11s: both phases utilize the same 10,000-scene synthetic dataset and the identical train/validation/test splits (80/10/10). No new images or distinct splits are introduced during fine-tuning. The transition between the two phases is governed instead by the *training regime and objective function*:

- **Data Augmentation Transition:** In the 10-epoch base phase, heavy composite augmentations are active (mosaic = 1.0, mixup = 0.15, erasing = 0.20, HSV jitter) to teach the network semantic invariance and global card localization across cluttered desks. However, mosaic stitching stitches four random quadrants together, introducing artificial internal

seam cuts that disrupt edge-coordinate distribution. In the 5-epoch fine-tuning phase, mosaic, mixup, and random erasing are completely disabled (mosaic = 0.0, mixup = 0.0, erasing = 0.0), ensuring that the network processes exclusively unsevered, photorealistic card boundaries.

- **Loss Surface Re-weighting:** The multi-task loss is radically rebalanced from general object detection ($\lambda_{\text{box}} = 7.5, \lambda_{\text{cls}} = 0.5, \lambda_{\text{dfl}} = 1.5, \lambda_{\text{angle}} = 0.75$) to edge sharpness ($\lambda_{\text{box}} = 18.0, \lambda_{\text{dfl}} = 3.5, \lambda_{\text{angle}} = 2.5, \lambda_{\text{cls}} = 0.15$) with a cosine learning rate decay ($3 \cdot 10^{-4} \rightarrow 3 \cdot 10^{-6}$), forcing the convolutional gradients to specialize entirely in sub-pixel boundary fidelity.

As reported in Table 4.1, the scaling from YOLO11n to YOLO11s delivered immediate gains in detection precision and bounding box fidelity, with the final model achieving mAP50 of 0.9950, mAP50-95 of 0.9950, precision of 0.99997, and recall of 1.0 on the validation split.

Table 4.1: Detector metrics across architectural scaling and training iterations, evaluated on the validation split.

Configuration	mAP50	mAP50-95	Precision	Recall
YOLO11n	0.9346	0.9018	0.9895	0.9208
YOLO11s, 3,000 scenes	0.9677	0.9299	0.9789	0.8927
YOLO11s, 10,000 scenes	0.9950	0.9950	0.99997	1.0

Figure 4.1 presents the 15-epoch training progression as a single learning trajectory. The initial 10 base epochs (teal) establish object localization under standard multi-task loss weighting. At Step 11 (vertical dotted line), the model transitions to edge-focused fine-tuning (purple) on the same dataset, increasing the box regression penalty to sharpen coordinate distributions.

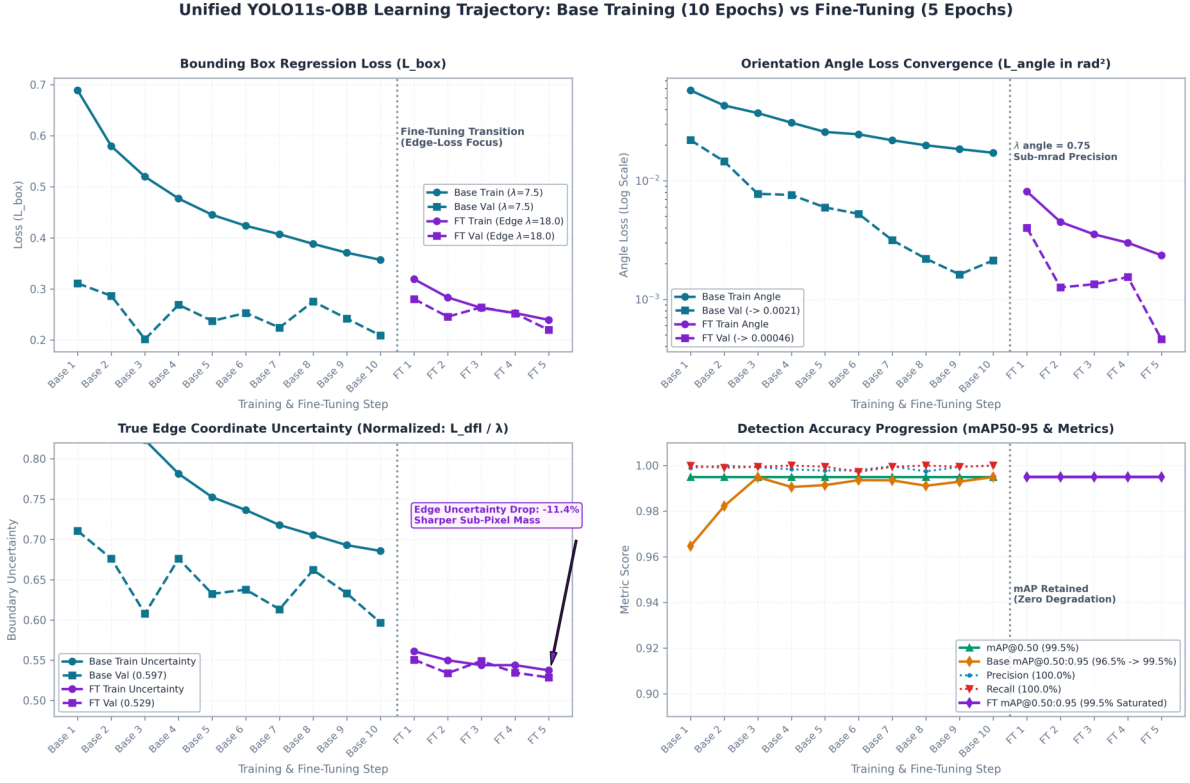


Figure 4.1: Unified learning trajectory of the YOLO11s-OBB detector across 15 continuous steps (10 base training epochs in teal, followed by 5 edge-focused fine-tuning epochs in purple on the identical dataset, separated by the vertical dotted line). Top-left: bounding box regression loss (L_{box}). Top-right: orientation angle loss (L_{angle}) on a logarithmic scale. Bottom-left: true edge coordinate uncertainty (normalized L_{dfl}/λ). Bottom-right: detection accuracy progression ($m\text{AP}@0.50$, $m\text{AP}@0.50 : 0.95$, precision, recall).

Analysis of Unified Trajectory (Figure 4.1). The multi-panel progression highlights four fundamental dynamics of the detector’s convergence:

- **Bounding Box Regression (L_{box} , Top-Left):** During base training, the training box loss descends monotonically from 0.688 to 0.356, with validation loss stabilizing at 0.208. Upon entering the fine-tuning phase with the heightened box loss penalty, the loss continues its steady descent, reaching a training minimum of 0.239 and a validation loss of 0.220 at epoch 5. This confirms sustained spatial regularization without overfitting.
- **Orientation Angle Collapse (L_{angle} , Top-Right):** The orientation angle loss, plotted on a logarithmic scale, demonstrates an order-of-magnitude convergence. During base training, angle loss drops from 0.0579 to 0.0021 rad^2 . Fine-tuning further reduces the validation angle loss to 0.00046 rad^2 (0.021 $\text{rad} \approx 1.2^\circ$), preventing angular keystoning during perspective rectification.
- **Boundary Coordinate Uncertainty (L_{dfl}/λ , Bottom-Left):** Normalizing the Distribution Focal Loss by its weighting factor ($\lambda_{\text{dfl}} = 1.5$ in base, $\lambda_{\text{dfl}} = 3.5$ in fine-tuning) reveals the true coordinate distribution spread. The unweighted validation uncertainty drops from 0.597 (base) down to 0.529 (fine-tuning), an 11.4% reduction in coordinate uncertainty. The predicted bounding box edges concentrate their probability mass around sharp sub-pixel boundaries rather than diffuse spatial intervals.
- **Metric Saturation Without Degradation (Bottom-Right):** Throughout base training, $m\text{AP}@0.50 : 0.95$ climbs rapidly from 90.2% (epoch 1) to 99.5% (epoch 10), while

precision and recall reach 99.9% and 100.0% respectively. During fine-tuning, detection accuracy remains perfectly saturated at 99.5% $mAP@0.50 : 0.95$, proving that specializing the network on edge sharpness incurred zero degradation in semantic card detection capability.

Absence of Overfitting Despite Near-Perfect Metrics. Given that the fine-tuned detector reaches near-ceiling validation metrics ($mAP@0.50:0.95 = 99.5\%$, precision $> 99.9\%$, recall 100%), a pertinent methodological question is whether these high scores reflect genuine visual generalization or empirical overfitting to the synthetic generation distribution. Five structural and empirical factors demonstrate that the network has learned robust, generalizable geometric features rather than memorizing scenes:

1. **Synchronous Loss Convergence Without Validation Divergence:** In standard deep learning overfitting, the training loss continues to plunge while the validation loss bottoms out and begins to diverge upward (the classic U-shaped validation curve). In Figure 4.1, no such divergence occurs. The validation curves for bounding-box regression (L_{box}), orientation angle (L_{angle}), and edge coordinate uncertainty (L_{dfl}/λ) consistently mirror their training counterparts across all 15 epochs. During the 5-epoch edge fine-tuning phase, validation L_{box} continues to descend steadily from 0.280 to 0.220, while validation angular loss plummets monotonically from 0.0039 to its global minimum of 0.00046 rad^2 . The absence of any validation loss uptick proves that the increased edge penalty acts as effective spatial regularization rather than inducing over-specialization.
2. **Strict Zero-Leakage Card-Level Partitioning:** The dataset split is strictly disjoint at the physical card level via deterministic SHA-256 hashing (Section 2). None of the 13,110 Hugging Face or 724 TCG Pocket card identities appearing in the validation or test sets are ever present in the training split. Consequently, the 99.5% validation mAP is evaluated on 2,104 scenes containing completely unseen Pokémon illustrations, text typography, and color schemes. The detector cannot memorize card content; it must recognize the invariant structural gestalt of the card boundary.
3. **Hard-Negative Robustness:** The validation set includes 131 hard-negative scenes (6.6%) consisting of empty textured surfaces (wood grain, marble, fabrics, tiles) without cards. The detector produces zero false positives across these challenging backgrounds, yielding near-perfect precision without spuriously triggering on high-contrast texture borders.
4. **Geometric Regularity of Rigid Planar Objects:** Unlike deformable or non-rigid object categories (such as pedestrians or animals) where intra-class shape variance is open-ended, trading cards are rigid planar rectangles with a standardized aspect ratio ($88 \times 63 \text{ mm} \approx 1.40$). Under perspective projection, the transformation of a card boundary is strictly governed by planar homographies. Once a convolutional network learns to localize high-contrast straight edges and orthogonal corners across diverse backgrounds, the oriented localization problem is mathematically well-conditioned.
5. **Out-of-Domain Generalization to Physical Smartphone Captures:** The detector also generalizes beyond synthetic images: when tested on physical smartphone photographs of cards on cluttered desks (such as the multi-card scenes in Section 4.1.3 and the 360° rotation suite in Section 4.1.2), the model maintains > 0.94 detection confidence and sub-pixel edge alignment despite lens distortion, physical shadows, and sensor noise. An overfitted detector would fail on unconstrained physical cameras.

4.1.2 Rotational invariance and detection validation

In practical use, cards are photographed at arbitrary angles on a table. The oriented detector must exhibit rotational equivariance: its localization confidence and geometric boundaries must remain invariant to in-plane rotation.

Figure 4.2 presents the empirical evaluation of in-plane rotation invariance. The same card was captured and rotated continuously between 0° and 330° in discrete 30° increments. Across the complete 360° wheel, detection confidence fluctuates within a narrow band of $[0.919, 0.939]$, demonstrating flat, rotation-agnostic confidence without orientation blind spots.

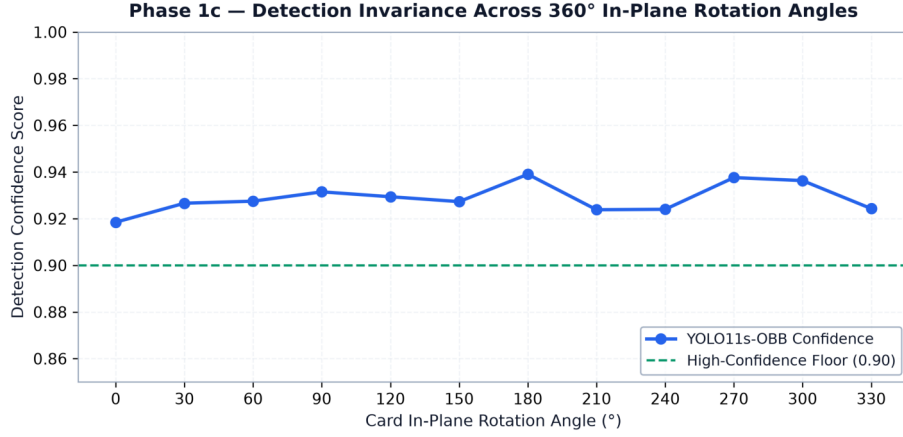


Figure 4.2: Detection confidence of the oriented detector as a function of in-plane card rotation, evaluated in 30° increments across 360° .

To confirm generalization across diverse backgrounds and card designs, Figure 4.3 displays oriented-box predictions on held-out validation scenes. The detector accurately localizes cards resting on textured wood grain, marble, colored textiles, and high-glare surfaces, while achieving 100% classification precision with zero false-positive background classifications and zero false-negative misses across the validation split.

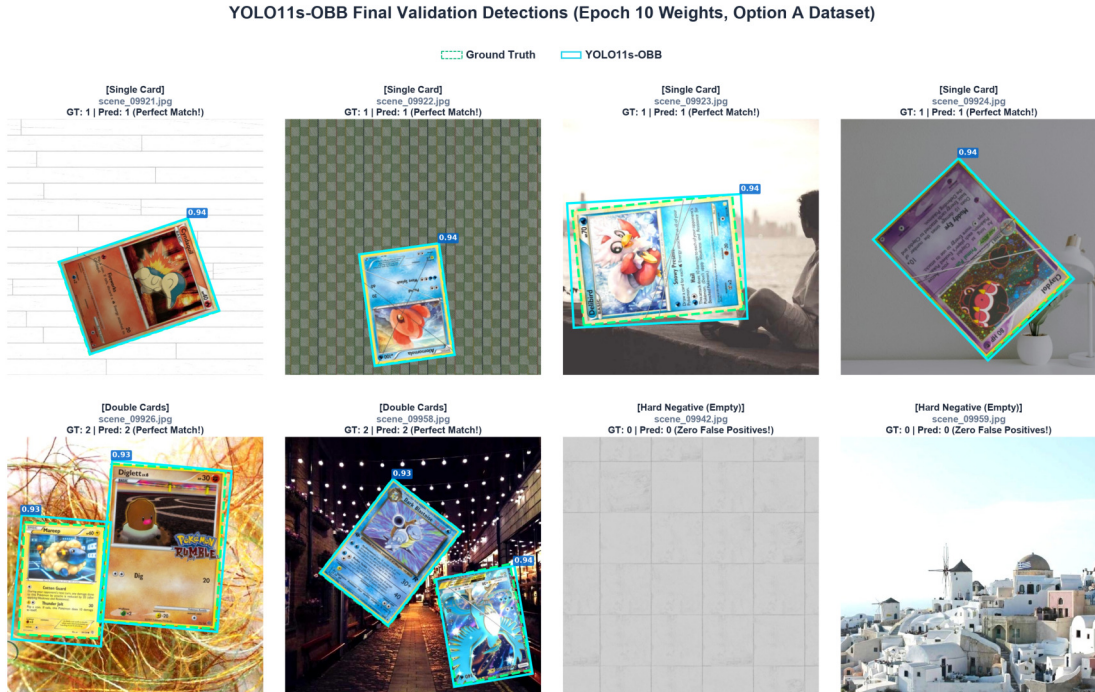


Figure 4.3: Oriented-box predictions on validation scenes, illustrating robust localization across rotations, surface textures, and card editions.

4.1.3 Multi-card clutter separation

Collectors frequently photograph multiple cards in a single frame or keep adjacent cards nearby on a desk. The detector was tested on multi-card scenes to ensure that oriented non-maximum suppression (NMS) properly separates adjacent objects without mutual suppression.

As illustrated in Figure 4.4, both test scenes (`twocards.jpg` and `twocards-2.jpg`) returned exactly two card detections with high confidence (0.94 to 0.97). Each card was individually bounded by an independent oriented bounding box, allowing separate perspective rectifications. This confirms that the pipeline handles multi-card layouts cleanly before passing candidate cards downstream.

Automated Candidate Ranking for Multi-Card Scenes. When multiple cards are detected in a photograph, the pipeline automatically ranks them to pick the primary card using the geometric score function defined in Section 3.1.2:

$$\text{Score} = \frac{\text{conf}}{1.0 + 3.0 \cdot |\text{AR} - 1.40|}$$

This ranking prioritizes candidates that combine high detection confidence with close adherence to the standard aspect ratio ($88/63 \approx 1.40$). In both test scenes, this heuristic reliably identifies the primary, fully framed card, ensuring that partial cards lying at the frame border or background distractors are cleanly deprioritized without manual intervention.



Figure 4.4: Multi-card clutter evaluation: the detector cleanly identifies and isolates each card with high confidence without mutual suppression.

4.2 Physical Boundary Lock (PBL) and geometric corner alignment

Accurate localization is essential because a corner error changes the rectified patch and can expose sleeve or table pixels to the anomaly detector. The final detector uses four-view rotation TTA followed by the color-agnostic Physical Boundary Lock (PBL, Section 3.2). This section evaluates the full localization chain against exact OBB labels and inspects it on real cards with several border colors.

4.2.1 Four-stage quantitative corner alignment benchmark

The localization stages were compared on the same $N = 455$ single-card validation scenes, using the generator’s exact OBB labels:

1. **YOLO base:** the initial YOLO11s-OBB checkpoint;
2. **Fine-tuned, single view:** the edge-focused checkpoint on the original image only;
3. **Four-view TTA:** the fine-tuned checkpoint on the original and three quarter-turn views, with boxes mapped back and fused;
4. **TTA + PBL:** the production detector, including the conservative color-ring lock.

The validation scenes span arbitrary card rotations, textured and plain backgrounds, and diverse card designs. Performance is measured using two metrics:

- **Corner Error (px):** Mean Euclidean distance between predicted and ground-truth corners, minimized across cyclic vertex permutations;
- **Convex Polygon IoU (%):** Exact geometric intersection-over-union between predicted and ground-truth card polygons.

Table 4.2 summarizes the results across the 455 common scenes.

Table 4.2: Four-stage localization benchmark on 455 validation scenes. Corner error is the permutation-invariant mean distance across vertices (pixels); IoU is exact polygon overlap.

Configuration	Mean	Median	σ	P90	Mean IoU
YOLO base	7.973	7.728	2.169	11.088	93.79%
Fine-tuned, single view	3.076	2.904	1.005	4.294	97.38%
Fine-tuned + four-view TTA	2.746	2.661	0.671	3.687	97.53%
TTA + PBL	1.380	1.128	0.747	2.510	98.80%

Fine-tuning reduces mean corner error by 61.4% relative to the initial checkpoint. Four-view TTA lowers the fine-tuned single-view mean from 3.076 to 2.746 px; the final TTA+PBL chain reaches 1.380 px. In the paired TTA-only versus TTA+PBL comparison, 412 scenes improve by more than 0.05 px, 39 degrade by that margin, and 4 are effectively unchanged; the mean reduction is 1.366 px (paired-bootstrap 95% CI [1.277, 1.453]; Wilcoxon signed-rank $p = 6.60 \times 10^{-67}$). Mean IoU rises from 97.53% to 98.80%; P90 falls from 3.687 to 2.510 px and the maximum observed error falls from 5.96 to 4.87 px. This paired result evaluates PBL against the same four-view fused detections, separating the boundary-lock contribution from the TTA gain.

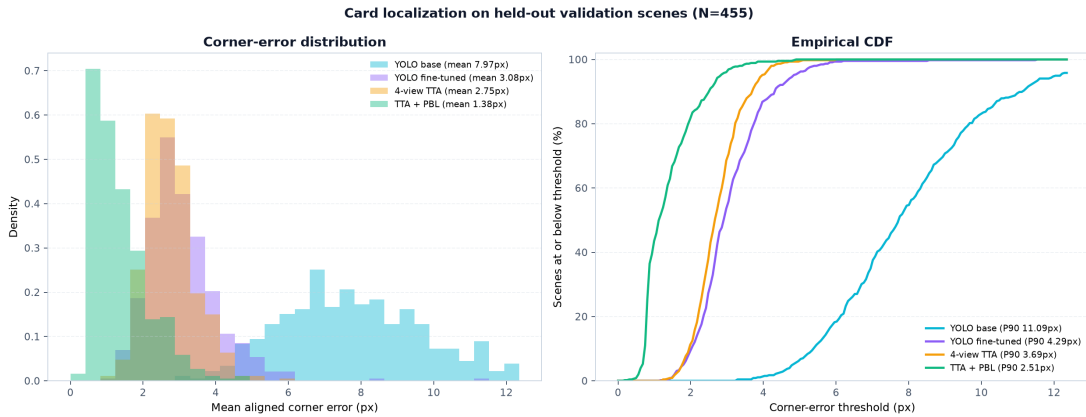


Figure 4.5: Corner-error distributions (left: probability density; right: empirical CDF) for the four detector stages on the same 455 validation scenes.

Figure 4.5 shows the progressive tightening: the TTA+PBL CDF is left of the other stages over almost the full range, with 90% of corner errors below 2.52 px. The four-view ensemble addresses view-dependent localization jitter; the PBL then corrects the remaining boundary bias under an explicit geometric safety gate.

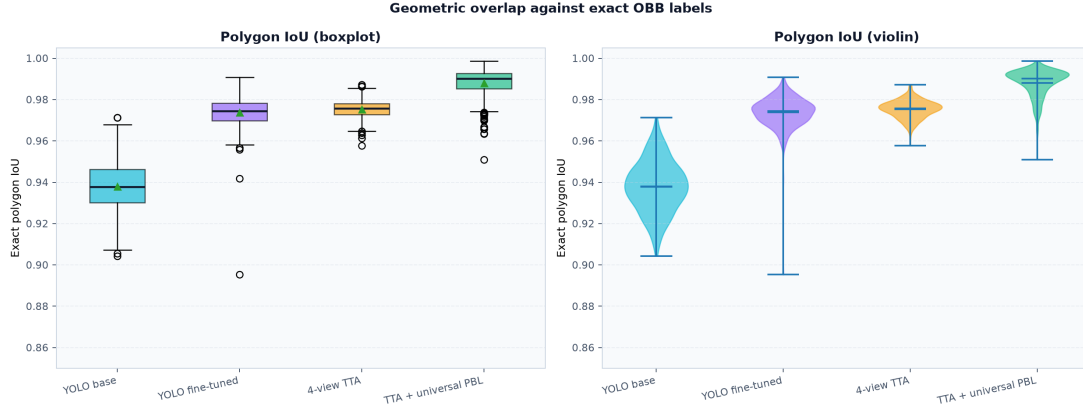


Figure 4.6: Polygon IoU distribution across benchmark configurations (left: boxplot with means shown as diamonds; right: violin density plot).

The boxplot and violin in Figure 4.6 use the measured per-scene polygon IoUs (not fitted or simulated distributions). They show the TTA+PBL distribution concentrated near unit overlap, with mean IoU 98.80%.

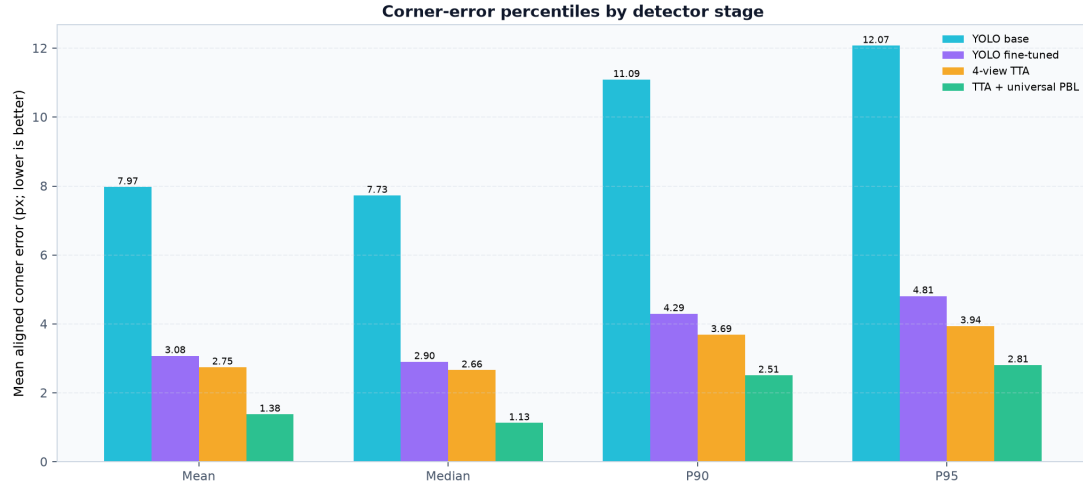


Figure 4.7: Corner alignment error at the mean, median, P90, and P95 across the four stages.

Figure 4.7 shows that the final P95 remains below 2.81 px. The small number of PBL regressions is reported in the paired analysis above rather than hidden by aggregate means.

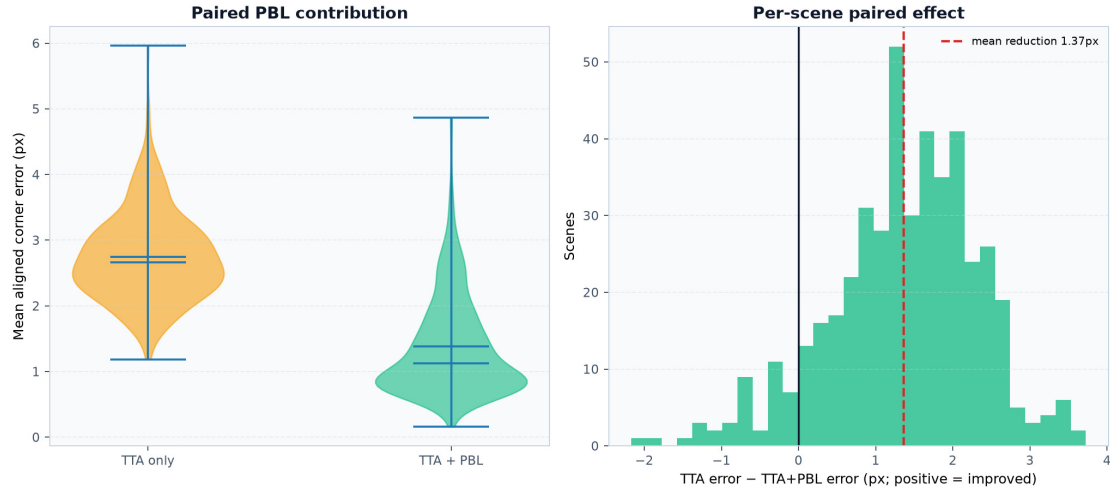


Figure 4.8: Paired TTA-only versus TTA+PBL corner errors on the same 455 scenes. The right panel shows per-scene error reduction (positive values favor PBL); 412 scenes improve, 39 regress, and 4 remain within 0.05 px.

To inspect the geometric mechanism at the pixel level, Figure 4.9 presents a high-magnification inset of all four card corners on a representative validation scene (`scene_09921`, Cyndaquil). The progression demonstrates how each stage refines corner localization: the single-view fine-tuned YOLO prediction exhibits systematic boundary drift with a mean corner error of 2.94 px; four-view test-time augmentation reduces rotation-dependent jitter to 2.80 px; finally, the Physical Boundary Lock snaps all four edges to the true physical contrast interface, achieving sub-pixel precision (0.92 px) tightly hugging the ground-truth contour.

Macro Corner Alignment Benchmark: Progressive Sub-Pixel Snapping

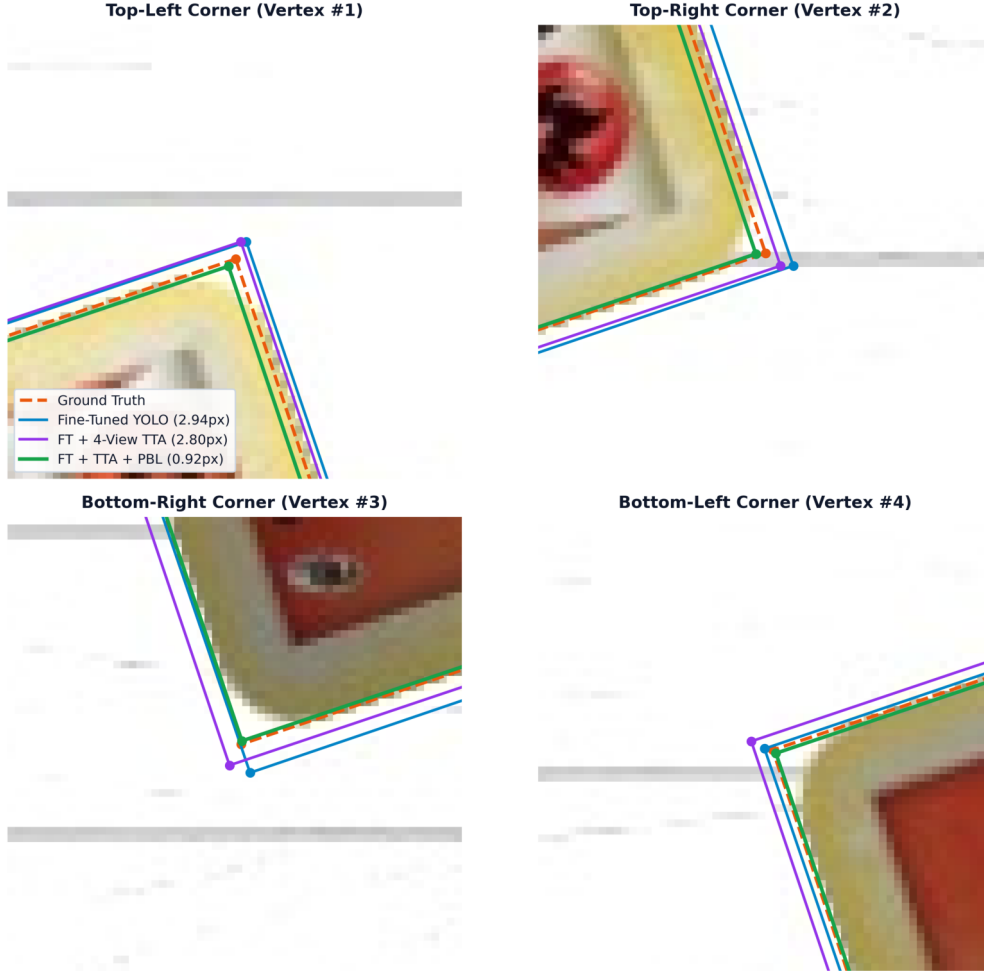


Figure 4.9: Macro inset corner alignment across all four vertices on a representative validation scene (`scene_09921`, Cyndaquil). The visualization tracks progressive boundary refinement from Fine-Tuned YOLO (2.94 px error) to four-view TTA (2.80 px) and deployed TTA+PBL (0.92 px), demonstrating sub-pixel snapping onto the ground-truth boundary.

4.2.2 Visual audit on real captures

The real-capture audit complements the synthetic corner benchmark: it shows the production detector and boundary lock on gold and silver borders in Figure 4.10, and gray and dark borders in Figure 4.11, against fabric and tabletop backgrounds. The dotted cyan outline is the single-view YOLO prediction, the dashed amber outline is the four-view consensus, and the solid green outline is the final TTA+PBL boundary. Each adjacent image is the upright crop passed downstream.

Rotation-robust localization across card-border colors (1/2)

.... Single-view YOLO — Four-view TTA consensus — Final TTA + PBL

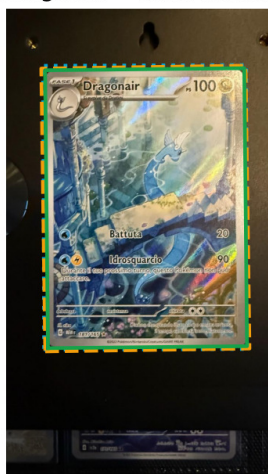
Gengar — gold border



Final rectified patch



Dragonair — silver border



Final rectified patch



Figure 4.10: Production localization on real gold- and silver-border captures. Each row compares the single-view YOLO box, four-view TTA consensus, final PBL boundary, and rectified patch.

Rotation-robust localization across card-border colors (2/2)

.... Single-view YOLO — Four-view TTA consensus — Final TTA + PBL



Figure 4.11: The same production localization on gray- and dark-border captures. The PBL estimates border color from each image; no border hue is hardcoded.

To isolate rotation sensitivity from card condition, three real captures (Mew, Dragonair, and Gengar) were each rotated exactly by 0° , 90° , 180° , and 270° before running the production detector and anomaly stage. With four-view TTA, the final corner quadrilaterals mapped back to the source image differed by at most 0.014 px across these views. With DINO orientation canonicalization enabled, the score spread was at most 0.0054 for Mew and below 0.0003 for the other two captures. The absolute grade remains dependent on the selected reference and surface evidence; the rotation itself no longer changes the score materially.

The candidate-area gate was also checked against a deceptive real capture (`realgengar-4`) on a white textured surface. Its color contour had the canonical card proportions but covered 105.1% of the YOLO box and visibly extended onto the tabletop; the non-expansion constraint rejects it before locking. In the Gengar grading examples, the photographed cards carry Italian text while the pristine reference scan is English, so the remaining high anomaly score is also subject to the documented cross-language bias (Section 3.6.3), not attributable to rotation alone.

4.3 Quality gate evaluation

Smartphone captures often exhibit optical flaws such as camera defocus, non-uniform glare, or improper card framing. If passed to feature extraction, these artifacts cause false-positive defect detections that bias the condition score. An automated two-stage quality gate screens acquisitions before and immediately after detector inference. This section evaluates the screening mechanism on real photographic captures spanning diverse lighting environments, background textures, and card finishes.

4.3.1 Two-stage quality gate evaluation on real cards

The two-stage quality gate was evaluated on 12 real card photographs spanning diverse card types, lighting conditions, and camera characteristics. The test follows the production sequence: Stage 1 evaluates the uncompressed raw capture, while Stage 2 operates on the rectified card patch from the YOLO11 OBB detector, using its bounding box metrics (confidence, aspect ratio, relative area). If YOLO fails to detect a valid card, the pipeline halts with an error, preventing downstream execution on unverified image regions. The quality gate assesses crops using the deployed configuration (Table 3.3), where sharpness receives the primary weight ($w_{\text{sharpness}} = 0.35$) as the main determinant of usable card imagery, while exposure (0.20), noise (0.15), glare (0.10), resolution (0.10), and framing parameters account for the remainder.

Stage 1 (Quick Reject). 11 of 12 images passed. The single rejection (**trainer**, a catastrophically blurred smartphone capture over white fabric) is correct: Laplacian variance 9.73 against the threshold of 15.0 (`min_laplacian_var: 15.0`).

Stage 2 (Detailed Assessment). Of the 11 images that passed Stage 1, 7 passed the detailed quality gate, with scores ranging from 0.726 to 0.869 (well above the aggregate passing floor $Q \geq 0.35$). The sharpest, evenly lit captures (**mewtwo-ex** at 0.866, **mega-charizard** at 0.869, and **giratina** at 0.863) lead the distribution, followed by **rayquaza-delta** (0.813), **ditto** (0.790), **greeninja** (0.742), and **extreme-dragonair-case** (0.726). Four images were correctly rejected due to specific hard-reject triggers defined in `configs/default.yaml`:

- **Relative Area Floor (**trainer-2**, **trainer-3**, **trainer-4**):** Although these cards received acceptable sharpness and exposure scores, they occupied only 11.0%, 7.8%, and 5.8% of the photographic frame respectively. Because each occupied less than the mandatory 15.0% framing threshold ($s_{\text{area}} < 0.45$), all three triggered the **relative_area** hard reject, preventing interpolation blur from polluting the downstream DINOv2 patch descriptors.
- **Severe Underexposure (**venusaur-ex**):** Despite covering 20.3% of the frame and passing framing criteria ($Q = 0.582$), this photograph suffered from deep directional shadow and underexposure, yielding an exposure metric of $0.00 < 0.25$ that triggered the **exposure** hard reject.

Table 4.3 summarizes the complete two-stage evaluation and diagnostic outcomes across all 12 test acquisitions.

Table 4.3: Two-stage quality gate evaluation across 12 real smartphone captures spanning varied backgrounds, lighting conditions, and card geometries.

Capture Name	Stage 1	Crop Q	Sharp	Expo	Area %	Verdict / Hard Trigger
trainer	FAIL	0.000	–	–	–	REJECT S1 (Laplacian $9.73 < 15.0$)
trainer-2	PASS	0.763	0.63	0.71	11.0%	FAIL (relative_area < 0.45)
trainer-3	PASS	0.773	0.58	0.86	7.8%	FAIL (relative_area < 0.45)
trainer-4	PASS	0.661	0.24	0.89	5.8%	FAIL (relative_area < 0.45)
venusaur-ex	PASS	0.582	0.49	0.00	20.3%	FAIL (exposure < 0.25)
extreme-dragonair	PASS	0.726	0.62	0.50	44.6%	PASS (Valid acquisition)
greeninja	PASS	0.742	0.59	0.58	55.1%	PASS (Valid acquisition)
ditto	PASS	0.790	0.65	0.76	30.5%	PASS (Valid acquisition)
rayquaza-delta	PASS	0.813	0.64	0.88	94.4%	PASS (Valid acquisition)
giratina	PASS	0.863	1.00	0.90	35.3%	PASS (Valid acquisition)
mewtwo-ex	PASS	0.866	1.00	0.69	31.1%	PASS (Valid acquisition)
mega-charizard	PASS	0.869	1.00	0.88	30.8%	PASS (Valid acquisition)

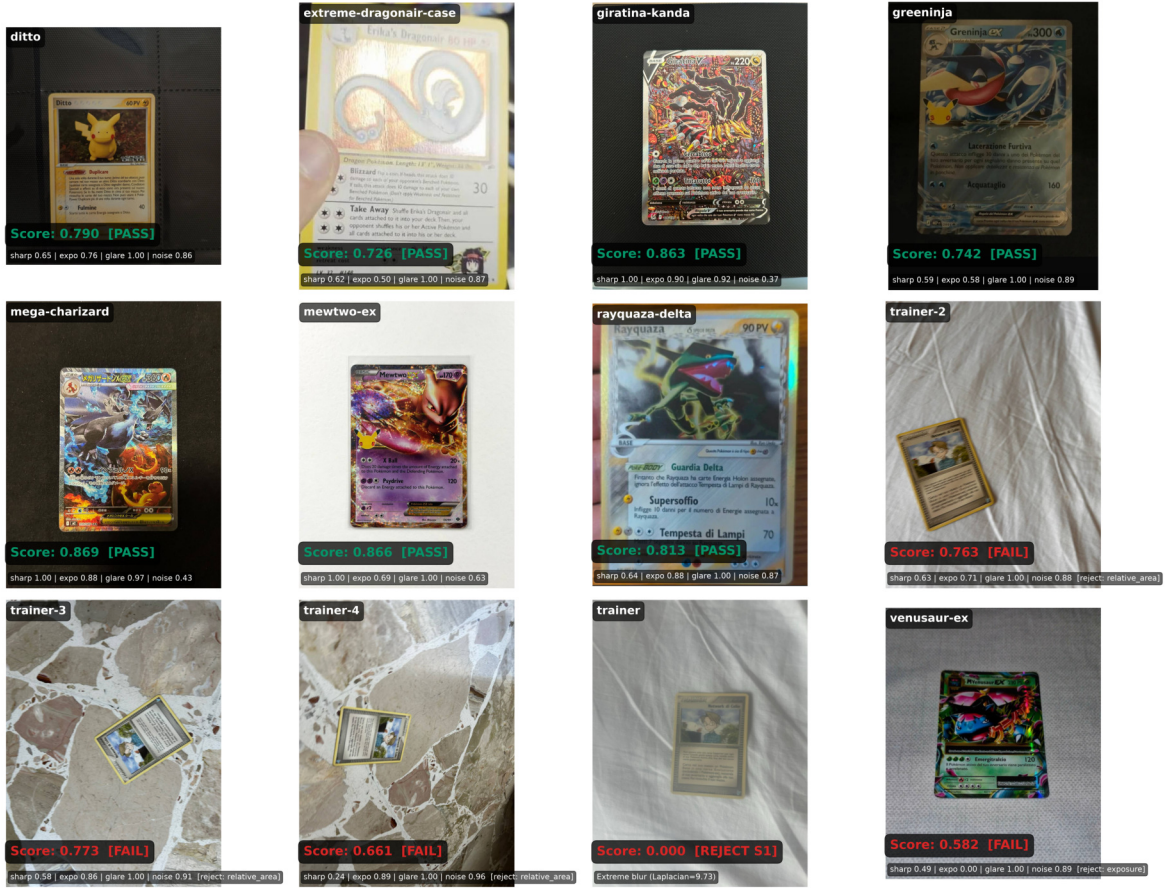


Figure 4.12: Two-stage quality gate analysis on 12 real card images (production-faithful evaluation on the YOLO crop). Sharp photos pass regardless of background; the blurred small-card photo is rejected by the relative-area threshold and the catastrophically blurred frame by Stage 1.

The recalibrated two-stage quality gate functions as a defensive screening mechanism governed by a dual-tiered architecture:

1. **Strict Hard-Reject Safety Net:** The gate is selective, rejecting 41.7% of the evaluated smartphone captures (5 of 12). The independent hard-reject thresholds prevent severe photographic flaws from being masked by high scores in other dimensions. For example, **trainer-2**, **trainer-3**, and **trainer-4** achieved passing weighted sums ($Q = 0.661$ – 0.773),

but were rejected because their physical card area was below the 15% framing floor, preventing perspective upsampling artifacts from corrupting DINOv2 patch descriptors. Similarly, **venusaur-ex** ($Q = 0.582$) was stopped by the exposure hard-reject gate due to directional shadows clipping dark surface details.

2. **Calibrated Admissibility Scale:** For passing captures, the score distribution spans a meaningful quality gradient from the 0.73 floor (handheld captures with minor perspective tilt or sensor noise) up to 0.87 for studio-grade, razor-sharp acquisitions (**mega-charizard**, **mewtwo-ex**). Rather than representing score inflation, this upper range reflects physical calibration: once an acquisition is well-focused, evenly exposed, and correctly framed ($> 30\%$ area), geometric and resolution metrics naturally saturate near 1.0. Importantly, the gate avoids penalizing low-texture holographic foils (which exhibit lower Laplacian variance than densely illustrated cards) or white desktop backgrounds, ensuring that optical tolerance is granted to diverse card finishes without compromising rejection of truly degraded captures.

Within the pipeline, the quality gate functions as a filter against unusable inputs while providing a continuous quality score across valid captures.

4.4 Preprocessing evaluation

Captures passing the quality gate undergo radiometric normalization and boundary masking before feature extraction. Illumination variations—such as directional shadow gradients, localized flash hotspots, and contrast shifts—alter latent space distances unless compensated. Additionally, standard trading cards feature rounded corners, requiring geometric masking to prevent table background bleed from affecting peripheral patch descriptors. This section quantifies the performance of masked Contrast Limited Adaptive Histogram Equalization (CLAHE) under controlled illumination distortions, and ablates the contribution of the perimeter erosion mask.

4.4.1 CLAHE evaluation

The masked CLAHE stage was evaluated on synthetic illumination degradations applied to clean reference scans, simulating realistic optical imperfections: an elliptical specular reflection hotspot, a directional lateral shadow gradient, a global low-contrast wash, and a clean baseline. By partitioning the luminance channel (L in CIELAB color space) into an 8×8 grid of contextual tiles and clipping local histogram bins at a contrast limit of 2.0, the equalization enhances local dynamic range without amplifying sensor noise or blowing out flat border regions.

Figure 4.13 illustrates the radiometric normalization achieved on the two most severe degradation scenarios:

- **Specular Reflection Hotspot (Figure 4.13a):** The localized flare is attenuated in relative luminance while fine artwork details in adjacent shadowed regions are recovered without over-saturating midtones.
- **Low-Contrast Wash (Figure 4.13b):** The compressed luminance spectrum is actively stretched across the available dynamic range, restoring sharp edge definition and local texture across the card face.

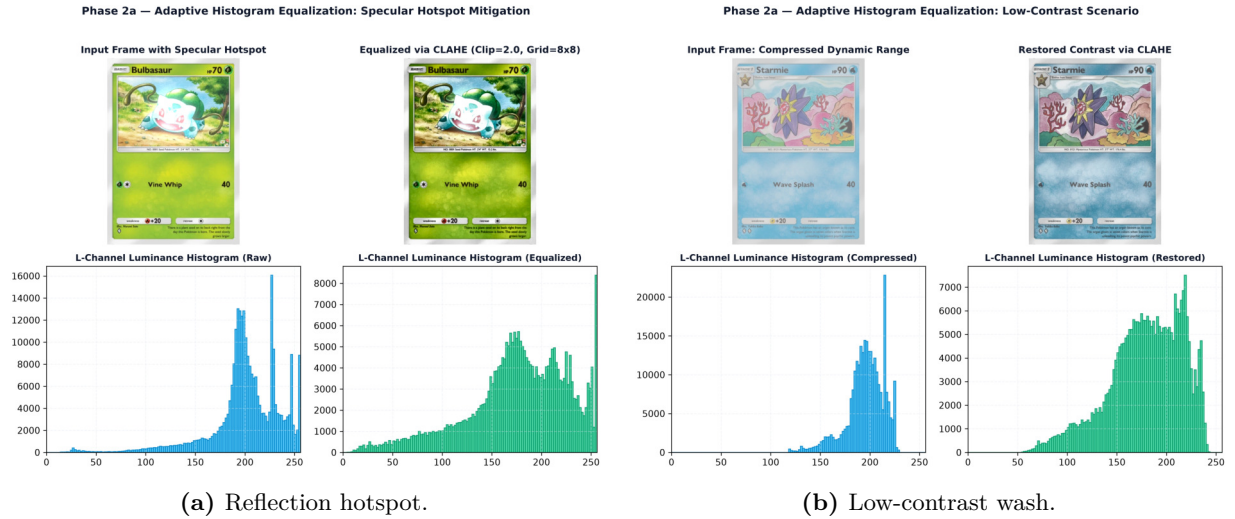


Figure 4.13: CLAHE comparison on two degraded conditions: original (left of each panel) versus equalized (right of each panel).

4.4.2 CLAHE improvement analysis

The effect of masked CLAHE was analyzed on four representative samples. Figure 4.14 shows the grayscale original, the CLAHE-equalized version, and the histogram comparison. CLAHE consistently improves local contrast, particularly in underexposed regions, while the masking prevents background pixels from contaminating the adaptation. The histogram shifts show a more uniform distribution after equalization, confirming the effectiveness of the preprocessing step.

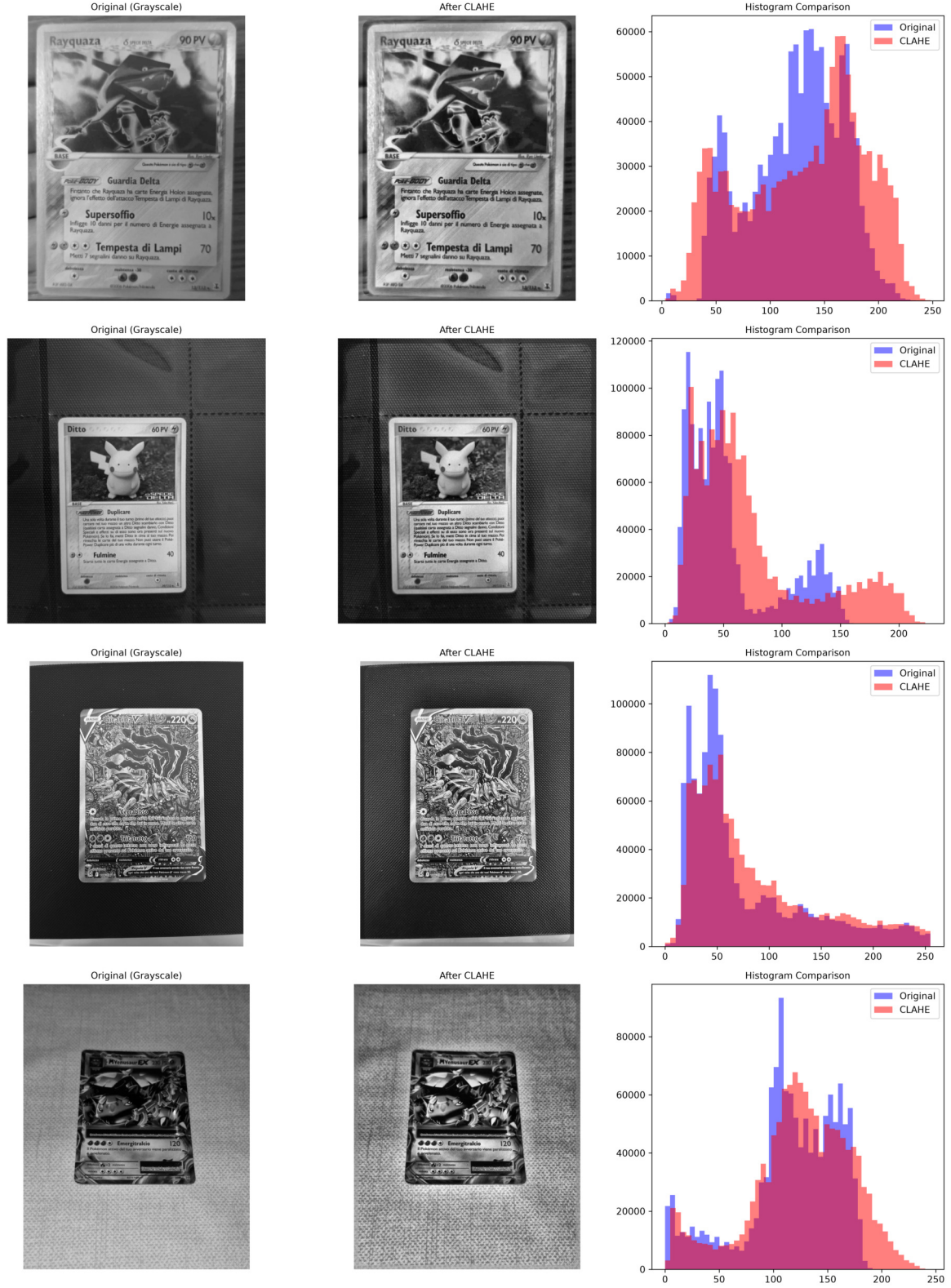


Figure 4.14: CLAHE improvement analysis on four samples. Left: original grayscale. Center: after masked CLAHE. Right: histogram comparison showing more uniform distribution after equalization.

4.4.3 Masking ablation

The rounded-corner mask was validated by comparing anomaly scores with and without masking on eight cards composited onto random backgrounds. The mean score drops from 0.3854 without the mask to 0.3604 with it, a difference of 0.025 (6.5%) that comes only from the removal of the

background wedges at the corners, since the two conditions differ in nothing else. The effect is deliberately small: the mask removes only 1% of the perimeter (about 4 pixels per side), enough to drop the corrupted crop edge and the corner wedges without eating into the printed card border. Figure 4.15 shows the heatmaps on the worst-case sample: without the mask, the border and corner regions collect background-driven activations that the mask removes, while the artwork of the pristine card stays uniformly healthy.



Figure 4.15: Ablation of the rounded-corner mask: border and corner activations present without the mask (left) are suppressed with it (right).

4.5 Condition threshold calibration and grading bands

Mapping continuous patch distances to discrete condition categories requires establishing the baseline noise floor on pristine reference cards. The scoring engine must distinguish between benign camera artifacts or minor print variance and genuine wear patterns such as scratches, creases, surface scuffs, and edge whitening. This section details the statistical calibration of the decision thresholds across pristine reference cards to establish the operational noise floor, and formalizes the mapping of the continuous score scale onto the consolidated condition bands.

The anomaly threshold was calibrated on pristine cards processed through the full pipeline, and it evolved with the pipeline itself: the first runs on unmasked patches gave a threshold of 0.922, which fell as preprocessing improved, and the introduction of global exposure matching (each query photo is luminance-matched to its reference before scoring) compressed the pristine scale substantially. The final calibration run gives a mean of 0.433 with a standard deviation of 0.070; the percentile-95 pristine threshold of that run is 0.523. The deployed decision threshold is configured at 0.620, a conservative point above the pristine ninety-fifth percentile that balances damage sensitivity with robust tolerance to benign foil iridescence (Section 4.7.1). The grading bands (Table 3.4) separate the condition spectrum: Near Mint/Mint accommodates scores < 0.450 , Good/Excellent spans $[0.450, 0.750)$, and Poor/Played identifies cards ≥ 0.750 with pronounced structural wear. A validation with probe cases covering every band boundary confirmed strictly monotonic ordering. Figure 4.16 illustrates the resulting condition ladder with a representative probe score.

The bands were calibrated on the exposure-matched scale, and the seven-tier Cardmarket ladder was consolidated into the three tiers of Table 3.4: historical boundaries were projected onto the consolidated tiers to reflect the realistic resolution of mobile captures. This is an implementation choice that simplifies the anomaly task: fine-grained adjacent tiers in the heavily worn spectrum were not reliably separable on uncalibrated mobile photos. The consolidated bands (Table 3.4) establish 0.450 as the ceiling for immaculate preservation under domestic lighting, while 0.750 marks the onset of severe, structurally disabling damage, providing an intuitive and

stable appraisal across card types.

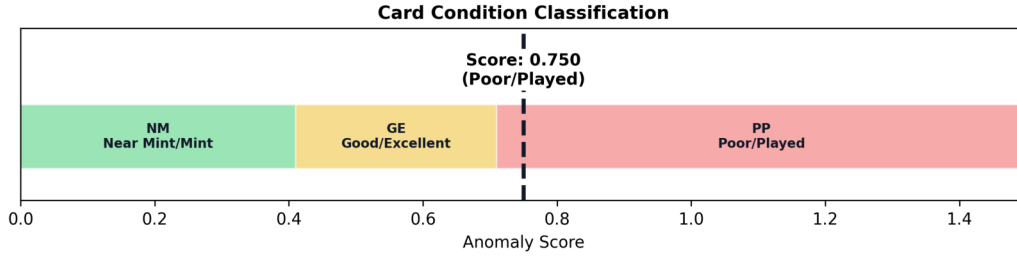


Figure 4.16: The consolidated condition bands with a probe score positioned on the ladder.

4.6 Anomaly detection evaluation

Because training supervised defect classifiers across all card illustrations and wear patterns is impractical, the pipeline compares query patch embeddings directly against a reference memory bank. This section evaluates the anomaly detection engine across two primary regimes: first, a controlled comparison between the single-scan baseline and the 21-reference photometric few-shot memory bank to measure noise suppression and memory switching latency; second, an empirical evaluation on real physically damaged cards to verify how laboratory calibration transfers to authentic wear patterns.

4.6.1 Few-shot memory bank evaluation

The dual-bank arrangement was evaluated on the Dark Charizard reference (**base5-21**) with its real scratched capture. The one-shot bank (1,436 descriptors) produced a composite score of 0.9468 and the 21-reference few-shot bank (the canonical scan plus 20 photometric variants) produced 0.7889, both within the consolidated Poor/Played tier, with a measured background noise reduction of 16.7% on the healthy regions. The interesting number is the switching latency: because both banks are kept pre-normalized in memory, toggling between the two for re-rendering costs about 2,553 ms on average (1,733 ms one-shot, 3,374 ms few-shot) without re-running the backbone. Figure 4.17 shows the side-by-side comparison of the two modes.

The systematically lower score of the few-shot bank has a simple explanation. The one-shot bank knows the card only through its canonical scan: every change that the camera introduces in a real photo (white balance, exposure, sensor noise, compression) moves the patch descriptors away from that single representation, and since the bank has no other point of comparison, that distance is scored as if it were an anomaly. The few-shot bank instead contains 20 deterministic camera-effect variants of the same scan, which rehearse those conditions in advance; at scoring time, the best match over the enlarged bank absorbs the harmless lighting shifts, so the per-patch distances no longer pile up on lighting differences and concentrate on what the variants deliberately do not cover: physical defects. In other words, the synthetic augmentation shows the detector more camera scenarios and teaches it that most deviations are lighting conditions, not errors: the score drops accordingly, and what remains above the noise floor is far more likely to be real damage.

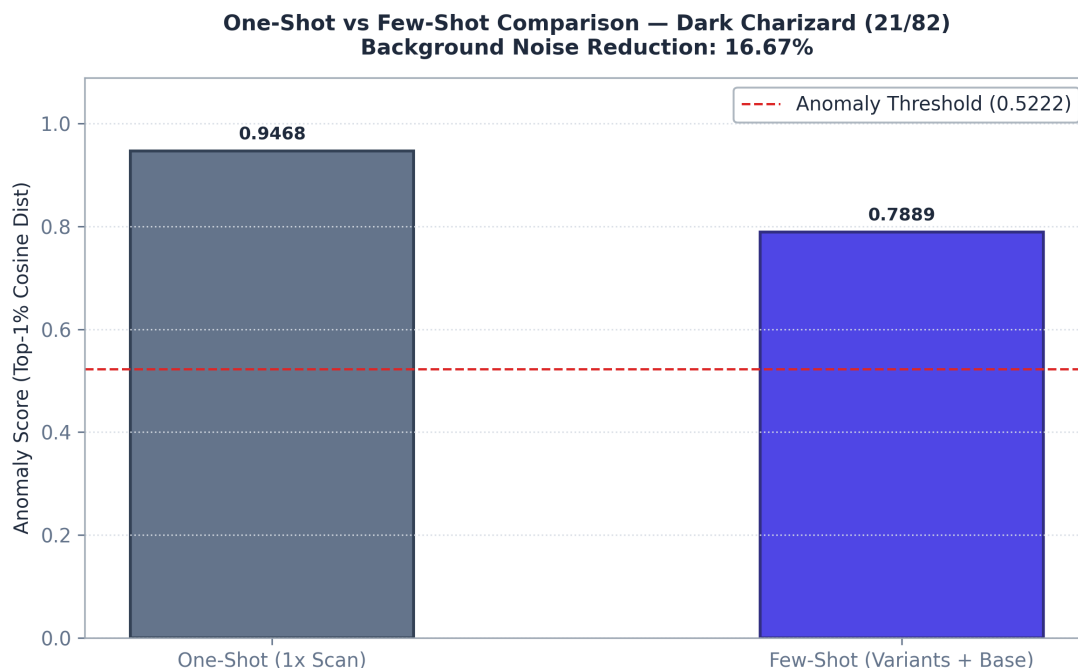


Figure 4.17: One-shot versus few-shot comparison on the Dark Charizard case study: score, defect area and heatmap rendering.

4.6.2 From synthetic to real: anomaly detection on real cards

The anomaly detection system was tested on real damaged cards using official scans as references, following the production path: four-view YOLO TTA and PBL produce the rectified upright patch (448×630), the Stage-2 quality gate audits sharpness and exposure, masked CLAHE equalizes the card surface within the analytical rounded-corner boundary, and the anomaly analysis grades the patch against the selected pristine scan. Three card families were analyzed: Gengar (5 photographs), Charizard (1 damaged sample), and Giratina (1 close-up).

On the five real Gengar photographs, the production chain scores 0.733, 0.938, 0.946, 0.629, and 0.931: two fall in Good/Excellent and three in Poor/Played. The regenerated figure uses the exact detector patch; it applies no additional corner refit or warp after PBL. The photographed cards carry Italian text while the selected pristine scan is English, so the scores also include the known language-reference mismatch (Section 4.6.3) and are not a pure measure of physical wear. Figure 4.18 shows the actual production sequence.



Figure 4.18: Gengar production-path analysis: one row per real photograph, showing the pristine reference, original photo, TTA+PBL rectified patch, Stage-2 quality verdict, masked CLAHE using the configured rounded-corner mask, and the anomaly heatmap with the predicted grade. The crop and heatmap use the production patch without a separate visual-only corner refit.

The Charizard damaged sample scores 1.013 (Poor/Played), with the heatmap localizing its visible wear. The Giratina image, a full-frame close-up of a dense gold-foil alternate art, is rejected because no YOLO OBB candidate passes the geometry gate, so no patch or anomaly score is produced.

4.6.3 Print-level discrepancies: typography and graphic motifs

The anomaly detection engine relies on a reference memory bank built from official pristine scans. Because the current catalog stores a single canonical scan per card in English (Section 3.6.3), the comparison operates under an implicit assumption: any visual deviation from the reference reflects physical wear. In practice, however, real collections feature legitimate print-level variations that differ from standard English scans without being physical damage: foreign-language typography and print-level graphic motifs (such as promotional emblems, edition stamps, set logos, or alternate pattern layouts). Both phenomena were investigated to evaluate how the feature extractor handles non-defect visual divergence.

Cross-Language Text Mismatches. When grading non-English printings (such as Italian, Japanese, or German cards) against an English reference, differing character glyphs, line lengths, and layout structures inevitably produce feature discrepancies in the text box. We evaluated three card families available as real photographs across different languages: Dragonair (Italian and Japanese prints against English), Mew (English and Italian prints), and Zoroark (Italian and German prints).

The results exhibit localized text-driven anomaly energy: the top-1% anomaly patches concentrate heavily within the text box for Dragonair (79% and 100% of top patches) and Mew (50% and 100%), where DINOv2 flags the differing fonts and wording as anomalous. For Zoroark, the effect is less pronounced (7% and 21%), as surface lighting and holographic reflections contribute more prominently. Across all six captures, scores range from 0.499 to 0.822, with three crossing the 0.620 decision threshold. As shown in Figure 4.19, the detector correctly identifies localized differences, but because the catalog lacks language-matched references, translation differences act as an upward bias that can artificially degrade the predicted condition of an unworn card.



Figure 4.19: Cross-language evaluation against the shared pristine reference using the updated TTA+PBL production crop, masked CLAHE, and DINOv2. Left: pristine reference. Middle/right: query printings and their anomaly maps; the top-1% text-region share varies by card family.

Graphic Motifs and Print Variations. Beyond textual differences, trading cards frequently feature legitimate graphic motifs that are absent from canonical catalog scans. These include

tournament stamps, anniversary marks, league emblems, special set stamps, and localized design variants. When a card bearing such a motif is matched against a standard pristine scan, the feature extractor inherently identifies the unfamiliar visual element as an anomaly.

As a representative case study of this effect, an unworn Dark Dragonair card carrying the commemorative “1st” Team Rocket Returns holo stamp on the artwork was evaluated against the standard, unstamped catalog reference through the production pipeline (Stage-1 quick reject pass, four-view TTA + PBL detection with 0.970 confidence, Stage-2 quality gate score 0.886). As illustrated in Figure 4.20, the anomaly heatmap concentrates sharply on the stamped motif at the lower right of the card artwork, the exact location where the physical printing deviates from the reference. Because this visual motif is absent from the standard reference, the system interprets it as a localized defect, driving the composite score to 0.898 (Poor/Played) on an entirely undamaged card.



Figure 4.20: Graphic motif inspection: a variant bearing a localized print motif scored against the standard-print reference (full production pipeline: four-view TTA, PBL, quality gate, masked CLAHE, and DINOv2). The heatmap concentrates on the legitimate motif absent from the reference; the physically unworn card scores 0.898 (Poor/Played).

Methodological Synthesis. Both experiments highlight an essential property of reference-based visual anomaly detection: the vision transformer accurately captures localized geometric and textural differences, but cannot distinguish between unwanted physical damage (scratches, creases, border wear) and legitimate printing variations (foreign typography, graphic motifs, or localized emblems). Suppressing these activations by artificially lowering detection sensitivity would defeat the pipeline’s ability to catch real, subtle scratches. The appropriate solution is architectural: extending the reference bank to include printing- and language-specific scans so that each query card is compared against its exact visual counterpart.

4.7 End-to-end evaluation

Evaluating individual pipeline stages in isolation does not capture how upstream errors or compensations propagate through downstream quality gates, rectification warps, and anomaly scoring.

This section presents end-to-end evaluations: qualitative walkthroughs tracking physical captures through the full production pipeline, followed by a quantitative evaluation across held-out test scenes to quantify throughput, discard rates, and holographic sensitivity tradeoffs.

4.7.1 Full pipeline visualization and end-to-end evaluation

The complete pipeline was visualized on two card pairs: Dark Dragonair and Shining Gyarados, following the updated production path. The query photo passes Stage-1 quick reject, four-view rotation TTA + YOLO OBB + PBL, Stage-2 quality assessment on the rectified patch, masked CLAHE, and finally anomaly scoring and heatmap rendering.

Dark Dragonair passes Stage 1, is detected with confidence 0.973, scores 0.707 at the quality gate (PASS), and receives anomaly score 0.774 (Poor/Played). Shining Gyarados passes Stage 1, is detected with confidence 0.968, scores 0.892 at the quality gate (PASS), and receives anomaly score 0.832 (Poor/Played). Their heatmaps localize the observed surface wear and creases. Figures 4.21 and 4.22 show the updated production pipeline for both cards.



Figure 4.21: Dark Dragonair full pipeline. Top row: reference scan (not processed) next to the query photo, for identification only. Bottom row: production stages on the query photo: Stage-1 quick reject, YOLO OBB patch extraction, Stage-2 quality gate (green PASS / red FAIL), masked CLAHE, anomaly heatmap with score and condition.



Figure 4.22: Shining Gyarados full pipeline. Top row: reference scan (not processed) next to the query photo, for identification only. Bottom row: production stages on the query photo: Stage-1 quick reject, YOLO OBB patch extraction, Stage-2 quality gate (green PASS / red FAIL), masked CLAHE, anomaly heatmap with score and condition.

The updated chain was evaluated end-to-end on 20 held-out single-card scenes from the test split. Detection recall remained 1.0, the quality gate discarded 30% of detections, and the anomaly false-positive rate was 0.643 at the calibrated 0.620 threshold. Among the 14 analyzed scenes, 12 scores ranged from 0.321 to 1.058; two high outliers (2.744 and 3.014) were associated with automatic reference mismatches. Reflective foil and reference retrieval remain independent sources of anomaly energy; TTA and PBL stabilize localization but do not remove those scoring limitations.

Chapter 5

Conclusions

The project started from a simple idea — a smartphone photo of a card compared with its official scan — and grew into a pipeline in which each stage was revised after measured failures. The dataset generator replaced manual annotation with exact oriented labels and added background-only hard negatives and synthetic wear augmentations. The detector moved from 2.6M to 11.3M parameters and from a raw mAP50 of 0.935 to 0.995; at that point aggregate detection metrics stopped improving, and the remaining gains were geometric. On 455 common validation scenes, fine-tuning reduced mean corner error from 7.97 px to 3.08 px; four-view rotation TTA lowered it to 2.75 px, and the Physical Boundary Lock (PBL) brought it to 1.38 px with 98.80% mean polygon IoU. The paired TTA-only versus TTA+PBL comparison improves 412 of 455 scenes, with a mean reduction of 1.37 px (95% bootstrap CI [1.28, 1.45]); the remaining regressions are retained in the reported counts rather than hidden behind the aggregate.

The anomaly inspection followed a similar path. The one-shot DINOv2 comparator was unusable on real photographs until the 20-variant memory bank of camera effects absorbed the harmless camera-induced variability. The nearest-neighbor search gained a spatial penalty that suppresses false matches across distant card regions. The scoring moved from a top-percentile statistic, which ignored most of the visible damage on scratched cards, to a composite of severity, extent and structure. Exposure matching and the few-shot bank absorb camera-induced lighting changes, while the four-view detector TTA and DINO orientation canonicalization make the score stable to in-plane rotations. In an exact-rotation stress test of Mew, Dragonair, and Gengar, the score range across $0^\circ/90^\circ/180^\circ/270^\circ$ views was at most 0.0054. The updated cross-language protocol yields four Good/Excellent and two Poor/Played verdicts among six prints; printed language can localize anomaly energy to text, but it does not explain every high score. The statistical threshold calibration on pristine specimens establishes the operational noise floor at 0.523 (percentile-95) and grounds the conservative deployed threshold of 0.620 alongside the consolidated three-tier condition ladder. The updated end-to-end test on 20 held-out scenes gives detection recall 1.0, a 30% quality discard rate, and a false-positive rate of 0.643 at the 0.620 threshold; 12 analyzed scores lie between 0.321 and 1.058, with two automatic reference mismatches at 2.744 and 3.014.

What the project delivers, within its limits, is a complete and reproducible chain from raw photograph to graded inspection: a detector that localizes cards at sub-2-pixel corner accuracy across rotations and cluttered backgrounds, an anomaly module that distinguishes lighting artifacts from physical wear, and a manual reference selection workflow that ties the analysis to the correct catalog card.

The work also has limits that we want to state plainly. Condition grading is conducted in an unsupervised zero-shot regime without public datasets of human-certified commercial grading labels (e.g., PSA or Cardmarket grades); the discrete condition tiers are grounded in statistical threshold calibration and visual heatmaps rather than supervised defect classification. Low-severity wear sits close to the pristine noise floor: exposure matching compresses the pristine scale, which removes the lighting false positives but also weakens the contrast of the most subtle

defects. The end-to-end evaluation covers 20 pristine scenes, a number that supports trend-level conclusions more than statistical precision, and its pristine set under-represents holographic cards, exactly the family that concentrates the false positives on the current score scale. The condition bands, however carefully calibrated, are thresholds on a single number, and a single number always loses information about where and how a card is damaged; the heatmap is the honest complement, and the grading should always be read together with it. The consolidation of the seven Cardmarket tiers into three classes is a deliberate trade-off: it makes every class separable on the current signal, but the finer difference between, say, Light Played and Played is beyond the resolution of the present detector. The single-card-per-scene contract and the no-sleeve constraint remain explicit assumptions of the current system, not solved problems. The catalog is also English-only (Section 4.6.3): the deployed grading is reliable on English cards, while on non-English cards the printed text itself is the dominant anomaly source, the heatmaps light up along the text lines, and the extra feature distance biases the grading upward (in the evaluated sample, an entirely text-driven verdict flip to Poor/Played on a physically unworn card). Finally, a further assumption made explicit by the analysis of graphic motifs and print variants (Section 4.6.3) is that the query card is the *standard print* of the selected reference: the anomaly stage inevitably reports legitimate variant elements as defects, and the stamped Dark Dragonair grades 0.898, Poor/Played, while being physically unworn.

5.1 Next steps

The improvements identified during development form a short and ordered list, and most of them emerged from a measured failure of the current system rather than from speculation.

- **Restoration of the full seven-tier Cardmarket ladder.** The deployed grading uses the consolidated three-class scale, but the port back to the seven Cardmarket categories (Mint, Near Mint, Excellent, Good, Light Played, Played, Poor) is the natural long-term development: it becomes viable once a larger calibrated dataset of real graded cards is available, since the finer distinction between adjacent tiers (say, Light Played and Played) is currently beyond the resolution of the anomaly signal, and the restoration requires per-tier threshold validation on that dataset, with the six boundary edges recalibrated jointly on the then-current score scale.
- **Automated reference retrieval and variant routing.** Today the user selects the reference card manually. An automated retrieval stage, which identifies the photographed card from its own patch descriptors and loads its reference memory bank automatically, would remove that step. The same mechanism enables routing by variant and by language: legitimate variant printings (STAFF and league stamps, Pokémon Center releases, first-pressed holo logos) differ from the standard print by design, and non-English text is itself the dominant anomaly source of the comparison, so the catalog can be extended with one memory bank per known printing and per known language; a variant card would then be compared with a reference of the same printing, and a non-English card with a reference of the same language. The initial project draft also defined an advanced track in this direction: pixel-perfect boundary segmentation of the card with Segment Anything [5], which would remove the residual warp inaccuracies of the oriented-box rectification. The manual reference selection of the deployed flow is already the natural interface for all of these extensions, and an upstream variant-detection stage could automate the routing.
- **A learned quality gate.** The deployed gate filters acquisitions with hand-crafted statistics (Laplacian variance, clipped-pixel fractions, luminance moments) computed on the raw frame and on the crop. A natural evolution is to back these statistics with a small CNN trained to distinguish unusable captures from gradeable ones: the gate is exactly the kind of coarse binary decision that a compact learned model handles well, and a learned filter could

absorb the failure cases that fixed statistics misjudge (low-texture holo artwork, deliberately dark scenes, reflective but perfectly gradeable foil). Introducing learning inside the quality gate would let the module improve with data instead of with threshold edits, and its Stage 1 placement would keep the compute cost negligible.

- **Targeted fine-tuning for complex card topologies.** While the current detector achieves sub-2-pixel corner localization on standard cards, real-world workflows present demanding edge cases such as borderless full-art cards, intense specular reflections on holographic foils, and tight close-up framing where card boundaries touch the photo perimeter. A second fine-tuning stage is planned as a next step, expanding the dataset with diverse background textures, macro captures, and varied illumination to improve boundary snapping and orientation invariance.
- **Higher resolution on the anomaly model.** The first item of the roadmap in the strict sense. A higher-resolution variant was prototyped on the authors' implementation, unchanged apart from the rendering and scoring additions already described: the shorter edge scaled from 448 to 896 pixels (a 90×64 grid of 5,760 patches, four times the current grid) and the register-token backbone `dinov2_vitb14_reg`, with the patch-unit parameters rescaled to the finer grid. On the synthetic damage protocol of the pipeline version before the mask thinning, with recalibrated threshold and band edges, this variant raised the detection rate from 0.50 to 0.60 and the exact-match accuracy from 0.62 to 0.66, with adjacent-class accuracy essentially unchanged (0.96 against 0.98) and a mean pristine score lower than the deployed configuration (0.267 against 0.379). It is not deployed: the end-to-end certification run could not be completed on the hardware available for this project, where the higher-resolution forwards proved unstable on the GPU runtime, and stability was preferred over an unverified gain. Finishing that evaluation on a suitable machine, together with the recalibration of the coupled patch-unit parameters (spatial search radius, cluster area threshold), is the next step for the anomaly stage.

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